

SOLVED PRACTICE WORKBOOK

Landforms by Running Water / India Drainage and Interlinking - Solved Practice Workbook

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing

Landforms by Running Water / India Drainage and Interlinking REFRESHED TEACHING NAVIGATION — BASIC/CORE FIRST	
01 FOUNDATION — BASIN, DIVIDE AND CHANNEL NETWORK A watershed is the boundary, not the entire river channel.	02 FOUNDATION — EROSION, TRANSPORT AND DEPOSITION Fluvial erosion detaches and wears material, transport moves
03 FOUNDATION — LONG PROFILE AND GRADED ADJUSTMENT Long-profile adjustment: A graded river tends toward a concave-up	04 FOUNDATION — UPPER-COURSE EROSIONAL SUITE Landforms reflect local gradient and resistance, not age labels alone.
05 CORE — MEANDERS, OX-BOWS AND FLOODPLAINS Meander and floodplain system: Helicoidal flow links outer-bank	06 CORE — DELTA VERSUS ESTUARY Sediment-budget comparison: Delta behaviour depends on basin
07 CORE — DRAINAGE PATTERNS AS STRUCTURAL EVIDENCE Drainage-pattern diagnosis: Dendritic drainage suggests broadly	08 CORE — ANTECEDENCE, SUPERIMPOSITION AND CAPTURE Evidence chain: Antecedence and superimposition - River capture
09 CORE — HIMALAYAN RIVER SYSTEMS Perennial does not mean uniform discharge or glacier-only flow.	10 CORE — PENINSULAR RIVER SYSTEMS Evidence chain: Peninsular drainage - Named river map hooks
11 CORE — FLOODPLAIN AND DELTA VULNERABILITY Rainfall alone does not explain disaster losses.	12 CORE — INTER-BASIN TRANSFER ACCOUNTING Inter-basin transfer test: A defensible link requires seasonal
13 SYNTHESIS — KEN-BETWA AND PANNA Evidence chain: Ken-Betwa direction and status - Panna landscape	14 SYNTHESIS — FLOOD, NAVIGATION AND FEDERAL LIMITS Evidence chain: Flood-navigation limits - Qualified basin governance
15 SYNTHESIS — VERIFIED ROUTES AND FINAL BASIN SPINE Evidence chain: Verified PYQ routes - Qualified basin governance	

Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

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BASIC MCQS / REMEDIATION

Q1. Which statement correctly explains Drainage basin anatomy?

A. A drainage basin is the area contributing runoff to a river system; a watershed or divide separates neighbouring basins, while tributaries join a trunk stream before its outlet. B. Running water erodes by hydraulic action, corrasion or abrasion, attrition and solution; it transports load by traction, saltation, suspension and solution before deposition when competence falls. C. A graded river tends toward a concave-up long profile in dynamic balance with discharge, sediment load, rock resistance and base level; uplift, floods and dams can interrupt the tendency. D. Steep gradients favour vertical incision, V-shaped valleys, gorges, rapids, waterfalls and interlocking spurs, but local rock and discharge matter more than a rigid stage label.

Answer: A. Explanation: A drainage basin is the area contributing runoff to a river system; a watershed or divide separates neighbouring basins, while tributaries join a trunk stream before its outlet. The other options describe different processes, locations, scales or governance categories.

Q2. Which option is the safest spatial interpretation of Drainage basin anatomy?

A. Steep gradients favour vertical incision, V-shaped valleys, gorges, rapids, waterfalls and interlocking spurs, but local rock and discharge matter more than a rigid stage label. B. A drainage basin is the area contributing runoff to a river system; a watershed or divide separates neighbouring basins, while tributaries join a trunk stream before its outlet. C. A graded river tends toward a concave-up long profile in dynamic balance with discharge, sediment load, rock resistance and base level; uplift, floods and dams can interrupt the tendency. D. Helicoidal flow links outer-bank erosion to inner-bank deposition; neck cutoff can form an ox-bow lake, while repeated overbank deposition builds floodplains and natural levees.

Answer: B. Explanation: A drainage basin is the area contributing runoff to a river system; a watershed or divide separates neighbouring basins, while tributaries join a trunk stream before its outlet. The other options describe different processes, locations, scales or governance categories.

Q3. Which statement preserves the process boundary for Drainage basin anatomy?

A. Steep gradients favour vertical incision, V-shaped valleys, gorges, rapids, waterfalls and interlocking spurs, but local rock and discharge matter more than a rigid stage label. B. A delta develops where sediment accumulation exceeds marine removal and subsidence, whereas an estuary is a drowned or tidally scoured river mouth; estuarine is not a delta shape. C. A drainage basin is the area contributing runoff to a river system; a watershed or divide separates neighbouring basins, while tributaries join a trunk stream before its outlet. D. Helicoidal flow links outer-bank erosion to inner-bank deposition; neck cutoff can form an ox-bow lake, while repeated overbank deposition builds floodplains and natural levees.

Answer: C. Explanation: A drainage basin is the area contributing runoff to a river system; a watershed or divide separates neighbouring basins, while tributaries join a trunk stream before its outlet. The other options describe different processes, locations, scales or governance categories.

Q4. Which option avoids the main UPSC trap concerning Drainage basin anatomy?

A. A delta develops where sediment accumulation exceeds marine removal and subsidence, whereas an estuary is a drowned or tidally scoured river mouth; estuarine is not a delta shape. B. Dendritic drainage suggests broadly uniform resistance; trellis alternating hard-soft bands; rectangular joints or faults; radial a central high; centripetal an enclosed basin. C. Helicoidal flow links outer-bank erosion to inner-bank deposition; neck cutoff can form an ox-bow lake, while repeated overbank deposition builds floodplains and natural levees. D. A drainage basin is the area contributing runoff to a river system; a watershed or divide separates neighbouring basins, while tributaries join a trunk stream before its outlet.

Answer: D. Explanation: A drainage basin is the area contributing runoff to a river system; a watershed or divide separates neighbouring basins, while tributaries join a trunk stream before its outlet. The other options describe different processes, locations, scales or governance categories.

Q5. Which statement correctly explains River work and load?

A. Running water erodes by hydraulic action, corrasion or abrasion, attrition and solution; it transports load by traction, saltation, suspension and solution before deposition when competence falls. B. Helicoidal flow links outer-bank erosion to inner-bank deposition; neck cutoff can form an ox-bow lake, while repeated overbank deposition builds floodplains and natural levees. C. A graded river tends toward a concave-up long profile in dynamic balance with discharge, sediment load, rock resistance and base level; uplift, floods and dams can interrupt the tendency. D. Steep gradients favour vertical incision, V-shaped valleys, gorges, rapids, waterfalls and interlocking spurs, but local rock and discharge matter more than a rigid stage label.

Answer: A. Explanation: Running water erodes by hydraulic action, corrasion or abrasion, attrition and solution; it transports load by traction, saltation, suspension and solution before deposition when competence falls. The other options describe different processes, locations, scales or governance categories.

Q6. Which option is the safest spatial interpretation of River work and load?

A. A delta develops where sediment accumulation exceeds marine removal and subsidence, whereas an estuary is a drowned or tidally scoured river mouth; estuarine is not a delta shape. B. Running water erodes by hydraulic action, corrasion or abrasion, attrition and solution; it transports load by traction, saltation, suspension and solution before deposition when competence falls. C. Helicoidal flow links outer-bank erosion to inner-bank deposition; neck cutoff can form an ox-bow lake, while repeated overbank deposition builds floodplains and natural levees. D. Steep gradients favour vertical incision, V-shaped valleys, gorges, rapids, waterfalls and interlocking spurs, but local rock and discharge matter more than a rigid stage label.

Answer: B. Explanation: Running water erodes by hydraulic action, corrasion or abrasion, attrition and solution; it transports load by traction, saltation, suspension and solution before deposition when competence falls. The other options describe different processes, locations, scales or governance categories.

Q7. Which statement preserves the process boundary for River work and load?

A. A delta develops where sediment accumulation exceeds marine removal and subsidence, whereas an estuary is a drowned or tidally scoured river mouth; estuarine is not a delta shape. B. Helicoidal flow links outer-bank erosion to inner-bank deposition; neck cutoff can form an ox-bow lake, while repeated overbank deposition builds floodplains and natural levees. C. Running water erodes by hydraulic action, corrasion or abrasion, attrition and solution; it transports load by traction, saltation, suspension and solution before deposition when competence falls. D. Dendritic drainage suggests broadly uniform resistance; trellis alternating hard-soft bands; rectangular joints or faults; radial a central high; centripetal an enclosed basin.

Answer: C. Explanation: Running water erodes by hydraulic action, corrasion or abrasion, attrition and solution; it transports load by traction, saltation, suspension and solution before deposition when competence falls. The other options describe different processes, locations, scales or governance categories.

Q8. Which option avoids the main UPSC trap concerning River work and load?

A. Dendritic drainage suggests broadly uniform resistance; trellis alternating hard-soft bands; rectangular joints or faults; radial a central high; centripetal an enclosed basin. B. An antecedent river maintains a course across rising relief by incision, while a superimposed river inherits a pattern from a removed cover and cuts into unrelated underlying structure. C. A delta develops where sediment accumulation exceeds marine removal and subsidence, whereas an estuary is a drowned or tidally scoured river mouth; estuarine is not a delta shape. D. Running water erodes by hydraulic action, corrasion or abrasion, attrition and solution; it transports load by traction, saltation, suspension and solution before deposition when competence falls.

Answer: D. Explanation: Running water erodes by hydraulic action, corrosion or abrasion, attrition and solution; it transports load by traction, saltation, suspension and solution before deposition when competence falls. The other options describe different processes, locations, scales or governance categories.

Q9. Which statement correctly explains Long-profile adjustment?

A. A graded river tends toward a concave-up long profile in dynamic balance with discharge, sediment load, rock resistance and base level; uplift, floods and dams can interrupt the tendency. B. A delta develops where sediment accumulation exceeds marine removal and subsidence, whereas an estuary is a drowned or tidally scoured river mouth; estuarine is not a delta shape. C. Helicoidal flow links outer-bank erosion to inner-bank deposition; neck cutoff can form an ox-bow lake, while repeated overbank deposition builds floodplains and natural levees. D. Steep gradients favour vertical incision, V-shaped valleys, gorges, rapids, waterfalls and interlocking spurs, but local rock and discharge matter more than a rigid stage label.

Answer: A. Explanation: A graded river tends toward a concave-up long profile in dynamic balance with discharge, sediment load, rock resistance and base level; uplift, floods and dams can interrupt the tendency. The other options describe different processes, locations, scales or governance categories.

Q10. Which option is the safest spatial interpretation of Long-profile adjustment?

A. Dendritic drainage suggests broadly uniform resistance; trellis alternating hard-soft bands; rectangular joints or faults; radial a central high; centripetal an enclosed basin. B. A graded river tends toward a concave-up long profile in dynamic balance with discharge, sediment load, rock resistance and base level; uplift, floods and dams can interrupt the tendency. C. Helicoidal flow links outer-bank erosion to inner-bank deposition; neck cutoff can form an ox-bow lake, while repeated overbank deposition builds floodplains and natural levees. D. A delta develops where sediment accumulation exceeds marine removal and subsidence, whereas an estuary is a drowned or tidally scoured river mouth; estuarine is not a delta shape.

Answer: B. Explanation: A graded river tends toward a concave-up long profile in dynamic balance with discharge, sediment load, rock resistance and base level; uplift, floods and dams can interrupt the tendency. The other options describe different processes, locations, scales or governance categories.

Q11. Which statement preserves the process boundary for Long-profile adjustment?

A. An antecedent river maintains a course across rising relief by incision, while a superimposed river inherits a pattern from a removed cover and cuts into unrelated underlying structure. B. Dendritic drainage suggests broadly uniform resistance; trellis alternating hard-soft bands; rectangular joints or faults; radial a central high; centripetal an enclosed basin. C. A graded river tends toward a concave-up long profile in dynamic balance with discharge, sediment load, rock resistance and base level; uplift, floods and dams can interrupt the tendency. D. A delta develops where sediment accumulation exceeds marine removal and subsidence, whereas an estuary is a drowned or tidally scoured river mouth; estuarine is not a delta shape.

Answer: C. Explanation: A graded river tends toward a concave-up long profile in dynamic balance with discharge, sediment load, rock resistance and base level; uplift, floods and dams can interrupt the tendency. The other options describe different processes, locations, scales or governance categories.

Q12. Which option avoids the main UPSC trap concerning Long-profile adjustment?

A. Dendritic drainage suggests broadly uniform resistance; trellis alternating hard-soft bands; rectangular joints or faults; radial a central high; centripetal an enclosed basin. B. An antecedent river maintains a course across rising relief by incision, while a superimposed river inherits a pattern from a removed cover and cuts into unrelated underlying structure. C. Headward erosion can divert a neighbouring stream, producing an elbow of capture, wind gap and misfit reach; capture is a drainage-evolution process, not ordinary tributary joining. D. A graded river tends toward a concave-up long profile in dynamic balance with discharge, sediment load, rock resistance and base level; uplift, floods and dams can interrupt the tendency.

Answer: D. Explanation: A graded river tends toward a concave-up long profile in dynamic balance with discharge, sediment load, rock resistance and base level; uplift, floods and dams can interrupt the tendency. The other options describe different processes, locations, scales or governance categories.

Q13. Which statement correctly explains Upper-course suite?

A. Steep gradients favour vertical incision, V-shaped valleys, gorges, rapids, waterfalls and interlocking spurs, but local rock and discharge matter more than a rigid stage label. B. Dendritic drainage suggests broadly uniform resistance; trellis alternating hard-soft bands; rectangular joints or faults; radial a central high; centripetal an enclosed basin. C. Helicoidal flow links outer-bank erosion to inner-bank deposition; neck cutoff can form an ox-bow lake, while repeated overbank deposition builds floodplains and natural levees. D. A delta develops where sediment accumulation exceeds marine removal and subsidence, whereas an estuary is a drowned or tidally scoured river mouth; estuarine is not a delta shape.

Answer: A. Explanation: Steep gradients favour vertical incision, V-shaped valleys, gorges, rapids, waterfalls and interlocking spurs, but local rock and discharge matter more than a rigid stage label. The other options describe different processes, locations, scales or governance categories.

Q14. Which option is the safest spatial interpretation of Upper-course suite?

A. Dendritic drainage suggests broadly uniform resistance; trellis alternating hard-soft bands; rectangular joints or faults; radial a central high; centripetal an enclosed basin. B. Steep gradients favour vertical incision, V-shaped valleys, gorges, rapids, waterfalls and interlocking spurs, but local rock and discharge matter more than a rigid stage label. C. A delta develops where sediment accumulation exceeds marine removal and subsidence, whereas an estuary is a drowned or tidally scoured river mouth; estuarine is not a delta shape. D. An antecedent river maintains a course across rising relief by incision, while a superimposed river inherits a pattern from a removed cover and cuts into unrelated underlying structure.

Answer: B. Explanation: Steep gradients favour vertical incision, V-shaped valleys, gorges, rapids, waterfalls and interlocking spurs, but local rock and discharge matter more than a rigid stage label. The other options describe different processes, locations, scales or governance categories.

Q15. Which statement preserves the process boundary for Upper-course suite?

A. An antecedent river maintains a course across rising relief by incision, while a superimposed river inherits a pattern from a removed cover and cuts into unrelated underlying structure. B. Headward erosion can divert a neighbouring stream, producing an elbow of capture, wind gap and misfit reach; capture is a drainage-evolution process, not ordinary tributary joining. C. Steep gradients favour vertical incision, V-shaped valleys, gorges, rapids, waterfalls and interlocking spurs, but local rock and discharge matter more than a rigid stage label. D. Dendritic drainage suggests broadly uniform resistance; trellis alternating hard-soft bands; rectangular joints or faults; radial a central high; centripetal an enclosed basin.

Answer: C. Explanation: Steep gradients favour vertical incision, V-shaped valleys, gorges, rapids, waterfalls and interlocking spurs, but local rock and discharge matter more than a rigid stage label. The other options describe different processes, locations, scales or governance categories.

Q16. Which option avoids the main UPSC trap concerning Upper-course suite?

A. Headward erosion can divert a neighbouring stream, producing an elbow of capture, wind gap and misfit reach; capture is a drainage-evolution process, not ordinary tributary joining. B. The Indus, Ganga and Brahmaputra systems cross high relief and international boundaries; their regimes combine rain, snow or glacier melt and groundwater in varying proportions. C. An antecedent river maintains a course across rising relief by incision, while a superimposed river inherits a pattern from a removed cover and cuts into unrelated underlying structure. D. Steep gradients favour vertical incision, V-shaped valleys, gorges, rapids, waterfalls and interlocking spurs, but local rock and discharge matter more than a rigid stage label.

Answer: D. Explanation: Steep gradients favour vertical incision, V-shaped valleys, gorges, rapids, waterfalls and interlocking spurs, but local rock and discharge matter more than a rigid stage label. The other options describe different processes, locations, scales or governance categories.

Q17. Which statement correctly explains Meander and floodplain system?

A. Helicoidal flow links outer-bank erosion to inner-bank deposition; neck cutoff can form an ox-bow lake, while repeated overbank deposition builds floodplains and natural levees. B. A delta develops where sediment accumulation exceeds marine removal and subsidence, whereas an estuary is a drowned or tidally scoured river mouth; estuarine is not a delta shape. C. Dendritic drainage suggests broadly uniform resistance; trellis alternating hard-soft bands; rectangular joints or faults; radial a central high; centripetal an enclosed basin. D. An antecedent river maintains a course across rising relief by incision, while a superimposed river inherits a pattern from a removed cover and cuts into unrelated underlying structure.

Answer: A. Explanation: Helicoidal flow links outer-bank erosion to inner-bank deposition; neck cutoff can form an ox-bow lake, while repeated overbank deposition builds floodplains and natural levees. The other options describe different processes, locations, scales or governance categories.

Q18. Which option is the safest spatial interpretation of Meander and floodplain system?

A. Dendritic drainage suggests broadly uniform resistance; trellis alternating hard-soft bands; rectangular joints or faults; radial a central high; centripetal an enclosed basin. B. Helicoidal flow links outer-bank erosion to inner-bank deposition; neck cutoff can form an ox-bow lake, while repeated overbank deposition builds floodplains and natural levees. C. An antecedent river maintains a course across rising relief by incision, while a superimposed river inherits a pattern from a removed cover and cuts into unrelated underlying structure. D. Headward erosion can divert a neighbouring stream, producing an elbow of capture, wind gap and misfit reach; capture is a drainage-evolution process, not ordinary tributary joining.

Answer: B. Explanation: Helicoidal flow links outer-bank erosion to inner-bank deposition; neck cutoff can form an ox-bow lake, while repeated overbank deposition builds floodplains and natural levees. The other options describe different processes, locations, scales or governance categories.

Q19. Which statement preserves the process boundary for Meander and floodplain system?

A. Headward erosion can divert a neighbouring stream, producing an elbow of capture, wind gap and misfit reach; capture is a drainage-evolution process, not ordinary tributary joining. B. The Indus, Ganga and Brahmaputra systems cross high relief and international boundaries; their regimes combine rain, snow or glacier melt and groundwater in varying proportions. C. Helicoidal flow links outer-bank erosion to inner-bank deposition; neck cutoff can form an ox-bow lake, while repeated overbank deposition builds floodplains and natural levees. D. An antecedent river maintains a course across rising relief by incision, while a superimposed river inherits a pattern from a removed cover and cuts into unrelated underlying structure.

Answer: C. Explanation: Helicoidal flow links outer-bank erosion to inner-bank deposition; neck cutoff can form an ox-bow lake, while repeated overbank deposition builds floodplains and natural levees. The other options describe different processes, locations, scales or governance categories.

Q20. Which option avoids the main UPSC trap concerning Meander and floodplain system?

A. Headward erosion can divert a neighbouring stream, producing an elbow of capture, wind gap and misfit reach; capture is a drainage-evolution process, not ordinary tributary joining. B. Most large Peninsular rivers follow the eastward plateau slope toward Bay of Bengal deltas, while Narmada and Tapi use west-flowing structural troughs and many Ghats streams are short and steep. C. The Indus, Ganga and Brahmaputra systems cross high relief and international boundaries; their regimes combine rain, snow or glacier melt and groundwater in varying proportions. D. Helicoidal flow links outer-bank erosion to inner-bank deposition; neck cutoff can form an ox-bow lake, while repeated overbank deposition builds floodplains and natural levees.

Answer: D. Explanation: Helicoidal flow links outer-bank erosion to inner-bank deposition; neck cutoff can form an ox-bow lake, while repeated overbank deposition builds floodplains and natural levees. The other options describe different processes, locations, scales or governance categories.

Q21. Which statement correctly explains Delta-estuary discriminator?

A. A delta develops where sediment accumulation exceeds marine removal and subsidence, whereas an estuary is a drowned or tidally scoured river mouth; estuarine is not a delta shape. B. An antecedent river maintains a course across rising relief by incision, while a superimposed river inherits a pattern from a removed cover and cuts into unrelated underlying structure. C. Dendritic drainage suggests broadly uniform resistance; trellis alternating hard-soft bands; rectangular joints or faults; radial a central high; centripetal an enclosed basin. D. Headward erosion can divert a neighbouring stream, producing an elbow of capture, wind gap and misfit reach; capture is a drainage-evolution process, not ordinary tributary joining.

Answer: A. Explanation: A delta develops where sediment accumulation exceeds marine removal and subsidence, whereas an estuary is a drowned or tidally scoured river mouth; estuarine is not a delta shape. The other options describe different processes, locations, scales or governance categories.

Q22. Which option is the safest spatial interpretation of Delta-estuary discriminator?

A. An antecedent river maintains a course across rising relief by incision, while a superimposed river inherits a pattern from a removed cover and cuts into unrelated underlying structure. B. A delta develops where sediment accumulation exceeds marine removal and subsidence, whereas an estuary is a drowned or tidally scoured river mouth; estuarine is not a delta shape. C. The Indus, Ganga and Brahmaputra systems cross high relief and international boundaries; their regimes combine rain, snow or glacier melt and groundwater in varying proportions. D. Headward erosion can divert a neighbouring stream, producing an elbow of capture, wind gap and misfit reach; capture is a drainage-evolution process, not ordinary tributary joining.

Answer: B. Explanation: A delta develops where sediment accumulation exceeds marine removal and subsidence, whereas an estuary is a drowned or tidally scoured river mouth; estuarine is not a delta shape. The other options describe different processes, locations, scales or governance categories.

Q23. Which statement preserves the process boundary for Delta-estuary discriminator?

A. Headward erosion can divert a neighbouring stream, producing an elbow of capture, wind gap and misfit reach; capture is a drainage-evolution process, not ordinary tributary joining. B. Most large Peninsular rivers follow the eastward plateau slope toward Bay of Bengal deltas, while Narmada and Tapi use west-flowing structural troughs and many Ghats streams are short and steep. C. A delta develops where sediment accumulation exceeds marine removal and subsidence, whereas an estuary is a drowned or tidally scoured river mouth; estuarine is not a delta shape. D. The Indus, Ganga and Brahmaputra systems cross high relief and international boundaries; their regimes combine rain, snow or glacier melt and groundwater in varying proportions.

Answer: C. Explanation: A delta develops where sediment accumulation exceeds marine removal and subsidence, whereas an estuary is a drowned or tidally scoured river mouth; estuarine is not a delta shape. The other options describe different processes, locations, scales or governance categories.

Q24. Which option avoids the main UPSC trap concerning Delta-estuary discriminator?

A. Most large Peninsular rivers follow the eastward plateau slope toward Bay of Bengal deltas, while Narmada and Tapi use west-flowing structural troughs and many Ghats streams are short and steep. B. The Indus, Ganga and Brahmaputra systems cross high relief and international boundaries; their regimes combine rain, snow or glacier melt and groundwater in varying proportions. C. The named Ganga begins at Devprayag after the Bhagirathi-Alaknanda confluence; Godavari rises near Trimbakeshwar, Krishna near Mahabaleshwar, Kaveri at Talakaveri and Narmada at Amarkantak. D. A delta develops where sediment accumulation exceeds marine removal and subsidence, whereas an estuary is a drowned or tidally scoured river mouth; estuarine is not a delta shape.

Answer: D. Explanation: A delta develops where sediment accumulation exceeds marine removal and subsidence, whereas an estuary is a drowned or tidally scoured river mouth; estuarine is not a delta shape. The other options describe different processes, locations, scales or governance categories.

Q25. Which statement correctly explains Drainage-pattern diagnosis?

A. Dendritic drainage suggests broadly uniform resistance; trellis alternating hard-soft bands; rectangular joints or faults; radial a central high; centripetal an enclosed basin. B. The Indus, Ganga and Brahmaputra systems cross high relief and international boundaries; their regimes combine rain, snow or glacier melt and groundwater in varying proportions. C. An antecedent river maintains a course across rising relief by incision, while a superimposed river inherits a pattern from a removed cover and cuts into unrelated underlying structure. D. Headward erosion can divert a neighbouring stream, producing an elbow of capture, wind gap and misfit reach; capture is a drainage-evolution process, not ordinary tributary joining.

Answer: A. Explanation: Dendritic drainage suggests broadly uniform resistance; trellis alternating hard-soft bands; rectangular joints or faults; radial a central high; centripetal an enclosed basin. The other options describe different processes, locations, scales or governance categories.

Q26. Which option is the safest spatial interpretation of Drainage-pattern diagnosis?

A. Headward erosion can divert a neighbouring stream, producing an elbow of capture, wind gap and misfit reach; capture is a drainage-evolution process, not ordinary tributary joining. B. Dendritic drainage suggests broadly uniform resistance; trellis alternating hard-soft bands; rectangular joints or faults; radial a central high; centripetal an enclosed basin. C. Most large Peninsular rivers follow the eastward plateau slope toward Bay of Bengal deltas, while Narmada and Tapi use west-flowing structural troughs and many Ghats streams are short and steep. D. The Indus, Ganga and Brahmaputra systems cross high relief and international boundaries; their regimes combine rain, snow or glacier melt and groundwater in varying proportions.

Answer: B. Explanation: Dendritic drainage suggests broadly uniform resistance; trellis alternating hard-soft bands; rectangular joints or faults; radial a central high; centripetal an enclosed basin. The other options describe different processes, locations, scales or governance categories.

Q27. Which statement preserves the process boundary for Drainage-pattern diagnosis?

A. The named Ganga begins at Devprayag after the Bhagirathi-Alaknanda confluence; Godavari rises near Trimbakeshwar, Krishna near Mahabaleshwar, Kaveri at Talakaveri and Narmada at Amarkantak. B. The Indus, Ganga and Brahmaputra systems cross high relief and international boundaries; their regimes combine rain, snow or glacier melt and groundwater in varying proportions. C. Dendritic drainage suggests broadly uniform resistance; trellis alternating hard-soft bands; rectangular joints or faults; radial a central high; centripetal an enclosed basin. D. Most large Peninsular rivers follow the eastward plateau slope toward Bay of Bengal deltas, while Narmada and Tapi use west-flowing structural troughs and many Ghats streams are short and steep.

Answer: C. Explanation: Dendritic drainage suggests broadly uniform resistance; trellis alternating hard-soft bands; rectangular joints or faults; radial a central high; centripetal an enclosed basin. The other options describe different processes, locations, scales or governance categories.

Q28. Which option avoids the main UPSC trap concerning Drainage-pattern diagnosis?

A. Most large Peninsular rivers follow the eastward plateau slope toward Bay of Bengal deltas, while Narmada and Tapi use west-flowing structural troughs and many Ghats streams are short and steep. B. Delta behaviour depends on basin sediment supply, wave-tidal energy, subsidence, relative sea level and distributary behaviour; equal sea-level forcing can produce different trajectories. C. The named Ganga begins at Devprayag after the Bhagirathi-Alaknanda confluence; Godavari rises near Trimbakeshwar, Krishna near Mahabaleshwar, Kaveri at Talakaveri and Narmada at Amarkantak. D. Dendritic drainage suggests broadly uniform resistance; trellis alternating hard-soft bands; rectangular joints or faults; radial a central high; centripetal an enclosed basin.

Answer: D. Explanation: Dendritic drainage suggests broadly uniform resistance; trellis alternating hard-soft bands; rectangular joints or faults; radial a central high; centripetal an enclosed basin. The other options describe different processes, locations, scales or governance categories.

Q29. Which statement correctly explains Antecedence and superimposition?

A. An antecedent river maintains a course across rising relief by incision, while a superimposed river inherits a pattern from a removed cover and cuts into unrelated underlying structure. B. The Indus, Ganga and Brahmaputra systems cross high relief and international boundaries; their regimes combine rain, snow or glacier melt and groundwater in varying proportions. C. Most large Peninsular rivers follow the eastward plateau slope toward Bay of Bengal deltas, while Narmada and Tapi use west-flowing structural troughs and many Ghats streams are short and steep. D. Headward erosion can divert a neighbouring stream, producing an elbow of capture, wind gap and misfit reach; capture is a drainage-evolution process, not ordinary tributary joining.

Answer: A. Explanation: An antecedent river maintains a course across rising relief by incision, while a superimposed river inherits a pattern from a removed cover and cuts into unrelated underlying structure. The other options describe different processes, locations, scales or governance categories.

Q30. Which option is the safest spatial interpretation of Antecedence and superimposition?

A. The Indus, Ganga and Brahmaputra systems cross high relief and international boundaries; their regimes combine rain, snow or glacier melt and groundwater in varying proportions. B. An antecedent river maintains a course across rising relief by incision, while a superimposed river inherits a pattern from a removed cover and cuts into unrelated underlying structure. C. The named Ganga begins at Devprayag after the Bhagirathi-Alaknanda confluence; Godavari rises near Trimbakeshwar, Krishna near Mahabaleshwar, Kaveri at Talakaveri and Narmada at Amarkantak. D. Most large Peninsular rivers follow the eastward plateau slope toward Bay of Bengal deltas, while Narmada and Tapi use west-flowing structural troughs and many Ghats streams are short and steep.

Answer: B. Explanation: An antecedent river maintains a course across rising relief by incision, while a superimposed river inherits a pattern from a removed cover and cuts into unrelated underlying structure. The other options describe different processes, locations, scales or governance categories.

Q31. Which statement preserves the process boundary for Antecedence and superimposition?

A. Delta behaviour depends on basin sediment supply, wave-tidal energy, subsidence, relative sea level and distributary behaviour; equal sea-level forcing can produce different trajectories. B. Most large Peninsular rivers follow the eastward plateau slope toward Bay of Bengal deltas, while Narmada and Tapi use west-flowing structural troughs and many Ghats streams are short and steep. C. An antecedent river maintains a course across rising relief by incision, while a superimposed river inherits a pattern from a removed cover and cuts into unrelated underlying structure. D. The named Ganga begins at Devprayag after the Bhagirathi-Alaknanda confluence; Godavari rises near Trimbakeshwar, Krishna near Mahabaleshwar, Kaveri at Talakaveri and Narmada at Amarkantak.

Answer: C. Explanation: An antecedent river maintains a course across rising relief by incision, while a superimposed river inherits a pattern from a removed cover and cuts into unrelated underlying structure. The other options describe different processes, locations, scales or governance categories.

Q32. Which option avoids the main UPSC trap concerning Antecedence and superimposition?

A. The named Ganga begins at Devprayag after the Bhagirathi-Alaknanda confluence; Godavari rises near Trimbakeshwar, Krishna near Mahabaleshwar, Kaveri at Talakaveri and Narmada at Amarkantak. B. Channel aggradation, migration, floodplain occupation, embankment confinement and drainage obstruction convert a natural inundation process into differentiated exposure and disaster risk. C. Delta behaviour depends on basin sediment supply, wave-tidal energy, subsidence, relative sea level and distributary behaviour; equal sea-level forcing can produce different trajectories. D. An antecedent river maintains a course across rising relief by incision, while a superimposed river inherits a pattern from a removed cover and cuts into unrelated underlying structure.

Answer: D. Explanation: An antecedent river maintains a course across rising relief by incision, while a superimposed river inherits a pattern from a removed cover and cuts into unrelated underlying structure. The other options describe different processes, locations, scales or governance categories.

Q33. Which statement correctly explains River capture?

A. Headward erosion can divert a neighbouring stream, producing an elbow of capture, wind gap and misfit reach; capture is a drainage-evolution process, not ordinary tributary joining. B. The named Ganga begins at Devprayag after the Bhagirathi-Alaknanda confluence; Godavari rises near Trimbakeshwar, Krishna near Mahabaleshwar, Kaveri at Talakaveri and Narmada at Amarkantak. C. The Indus, Ganga and Brahmaputra systems cross high relief and international boundaries; their regimes combine rain, snow or glacier melt and groundwater in varying proportions. D. Most large Peninsular rivers follow the eastward plateau slope toward Bay of Bengal deltas, while Narmada and Tapi use west-flowing structural troughs and many Ghats streams are short and steep.

Answer: A. Explanation: Headward erosion can divert a neighbouring stream, producing an elbow of capture, wind gap and misfit reach; capture is a drainage-evolution process, not ordinary tributary joining. The other options describe different processes, locations, scales or governance categories.

Q34. Which option is the safest spatial interpretation of River capture?

A. Most large Peninsular rivers follow the eastward plateau slope toward Bay of Bengal deltas, while Narmada and Tapi use west-flowing structural troughs and many Ghats streams are short and steep. B. Headward erosion can divert a neighbouring stream, producing an elbow of capture, wind gap and misfit reach; capture is a drainage-evolution process, not ordinary tributary joining. C. The named Ganga begins at Devprayag after the Bhagirathi-Alaknanda confluence; Godavari rises near Trimbakeshwar, Krishna near Mahabaleshwar, Kaveri at Talakaveri and Narmada at Amarkantak. D. Delta behaviour depends on basin sediment supply, wave-tidal energy, subsidence, relative sea level and distributary behaviour; equal sea-level forcing can produce different trajectories.

Answer: B. Explanation: Headward erosion can divert a neighbouring stream, producing an elbow of capture, wind gap and misfit reach; capture is a drainage-evolution process, not ordinary tributary joining. The other options describe different processes, locations, scales or governance categories.

Q35. Which statement preserves the process boundary for River capture?

A. Delta behaviour depends on basin sediment supply, wave-tidal energy, subsidence, relative sea level and distributary behaviour; equal sea-level forcing can produce different trajectories. B. Channel aggradation, migration, floodplain occupation, embankment confinement and drainage obstruction convert a natural inundation process into differentiated exposure and disaster risk. C. Headward erosion can divert a neighbouring stream, producing an elbow of capture, wind gap and misfit reach; capture is a drainage-evolution process, not ordinary tributary joining. D. The named Ganga begins at Devprayag after the Bhagirathi-Alaknanda confluence; Godavari rises near Trimbakeshwar, Krishna near Mahabaleshwar, Kaveri at Talakaveri and Narmada at Amarkantak.

Answer: C. Explanation: Headward erosion can divert a neighbouring stream, producing an elbow of capture, wind gap and misfit reach; capture is a drainage-evolution process, not ordinary tributary joining. The other options describe different processes, locations, scales or governance categories.

Q36. Which option avoids the main UPSC trap concerning River capture?

A. Channel aggradation, migration, floodplain occupation, embankment confinement and drainage obstruction convert a natural inundation process into differentiated exposure and disaster risk. B. A defensible link requires seasonal donor-basin accounting, recipient demand and hydrogeological fit, downstream and environmental-flow needs, conveyance losses and viable local alternatives. C. Delta behaviour depends on basin sediment supply, wave-tidal energy, subsidence, relative sea level and distributary behaviour; equal sea-level forcing can produce different trajectories. D. Headward erosion can divert a neighbouring stream, producing an elbow of capture, wind gap and misfit reach; capture is a drainage-evolution process, not ordinary tributary joining.

Answer: D. Explanation: Headward erosion can divert a neighbouring stream, producing an elbow of capture, wind gap and misfit reach; capture is a drainage-evolution process, not ordinary tributary joining. The other options describe different processes, locations, scales or governance categories.

Q37. Which statement correctly explains Himalayan drainage?

A. The Indus, Ganga and Brahmaputra systems cross high relief and international boundaries; their regimes combine rain, snow or glacier melt and groundwater in varying proportions. B. Most large Peninsular rivers follow the eastward plateau slope toward Bay of Bengal deltas, while Narmada and Tapi use west-flowing structural troughs and many Ghats streams are short and steep. C. Delta behaviour depends on basin sediment supply, wave-tidal energy, subsidence, relative sea level and distributary behaviour; equal sea-level forcing can produce different trajectories. D. The named Ganga begins at Devprayag after the Bhagirathi-Alaknanda confluence; Godavari rises near Trimbakeshwar, Krishna near Mahabaleshwar, Kaveri at Talakaveri and Narmada at Amarkantak.

Answer: A. Explanation: The Indus, Ganga and Brahmaputra systems cross high relief and international boundaries; their regimes combine rain, snow or glacier melt and groundwater in varying proportions. The other options describe different processes, locations, scales or governance categories.

Q38. Which option is the safest spatial interpretation of Himalayan drainage?

A. The named Ganga begins at Devprayag after the Bhagirathi-Alaknanda confluence; Godavari rises near Trimbakeshwar, Krishna near Mahabaleshwar, Kaveri at Talakaveri and Narmada at Amarkantak. B. The Indus, Ganga and Brahmaputra systems cross high relief and international boundaries; their regimes combine rain, snow or glacier melt and groundwater in varying proportions. C. Delta behaviour depends on basin sediment supply, wave-tidal energy, subsidence, relative sea level and distributary behaviour; equal sea-level forcing can produce different trajectories. D. Channel aggradation, migration, floodplain occupation, embankment confinement and drainage obstruction convert a natural inundation process into differentiated exposure and disaster risk.

Answer: B. Explanation: The Indus, Ganga and Brahmaputra systems cross high relief and international boundaries; their regimes combine rain, snow or glacier melt and groundwater in varying proportions. The other options describe different processes, locations, scales or governance categories.

Q39. Which statement preserves the process boundary for Himalayan drainage?

A. A defensible link requires seasonal donor-basin accounting, recipient demand and hydrogeological fit, downstream and environmental-flow needs, conveyance losses and viable local alternatives. B. Channel aggradation, migration, floodplain occupation, embankment confinement and drainage obstruction convert a natural inundation process into differentiated exposure and disaster risk. C. The Indus, Ganga and Brahmaputra systems cross high relief and international boundaries; their regimes combine rain, snow or glacier melt and groundwater in varying proportions. D. Delta behaviour depends on basin sediment supply, wave-tidal energy, subsidence, relative sea level and distributary behaviour; equal sea-level forcing can produce different trajectories.

Answer: C. Explanation: The Indus, Ganga and Brahmaputra systems cross high relief and international boundaries; their regimes combine rain, snow or glacier melt and groundwater in varying proportions. The other options describe different processes, locations, scales or governance categories.

Q40. Which option avoids the main UPSC trap concerning Himalayan drainage?

A. A defensible link requires seasonal donor-basin accounting, recipient demand and hydrogeological fit, downstream and environmental-flow needs, conveyance losses and viable local alternatives. B. Official NWDA material identifies Ken-to-Betwa transfer and the project as the first National Perspective Plan link under implementation; implementation is not proof of delivered outcomes. C. Channel aggradation, migration, floodplain occupation, embankment confinement and drainage obstruction convert a natural inundation process into differentiated exposure and disaster risk. D. The Indus, Ganga and Brahmaputra systems cross high relief and international boundaries; their regimes combine rain, snow or glacier melt and groundwater in varying proportions.

Answer: D. Explanation: The Indus, Ganga and Brahmaputra systems cross high relief and international boundaries; their regimes combine rain, snow or glacier melt and groundwater in varying proportions. The other options describe different processes, locations, scales or governance categories.

Q41. Which statement correctly explains Peninsular drainage?

A. Most large Peninsular rivers follow the eastward plateau slope toward Bay of Bengal deltas, while Narmada and Tapi use west-flowing structural troughs and many Ghats streams are short and steep. B. The named Ganga begins at Devprayag after the Bhagirathi-Alaknanda confluence; Godavari rises near Trimbakeshwar, Krishna near Mahabaleshwar, Kaveri at Talakaveri and Narmada at Amarkantak. C. Channel aggradation, migration, floodplain occupation, embankment confinement and drainage obstruction convert a natural inundation process into differentiated exposure and disaster risk. D. Delta behaviour depends on basin sediment supply, wave-tidal energy, subsidence, relative sea level and distributary behaviour; equal sea-level forcing can produce different trajectories.

Answer: A. Explanation: Most large Peninsular rivers follow the eastward plateau slope toward Bay of Bengal deltas, while Narmada and Tapi use west-flowing structural troughs and many Ghats streams are short and steep. The other options describe different processes, locations, scales or governance categories.

Q42. Which option is the safest spatial interpretation of Peninsular drainage?

A. Channel aggradation, migration, floodplain occupation, embankment confinement and drainage obstruction convert a natural inundation process into differentiated exposure and disaster risk. B. Most large Peninsular rivers follow the eastward plateau slope toward Bay of Bengal deltas, while Narmada and Tapi use west-flowing structural troughs and many Ghats streams are short and steep. C. A defensible link requires seasonal donor-basin accounting, recipient demand and hydrogeological fit, downstream and environmental-flow needs, conveyance losses and viable local alternatives. D. Delta behaviour depends on basin sediment supply, wave-tidal energy, subsidence, relative sea level and distributary behaviour; equal sea-level forcing can produce different trajectories.

Answer: B. Explanation: Most large Peninsular rivers follow the eastward plateau slope toward Bay of Bengal deltas, while Narmada and Tapi use west-flowing structural troughs and many Ghats streams are short and steep. The other options describe different processes, locations, scales or governance categories.

Q43. Which statement preserves the process boundary for Peninsular drainage?

A. A defensible link requires seasonal donor-basin accounting, recipient demand and hydrogeological fit, downstream and environmental-flow needs, conveyance losses and viable local alternatives. B. Channel aggradation, migration, floodplain occupation, embankment confinement and drainage obstruction convert a natural inundation process into differentiated exposure and disaster risk. C. Most large Peninsular rivers follow the eastward plateau slope toward Bay of Bengal deltas, while Narmada and Tapi use west-flowing structural troughs and many Ghats streams are short and steep. D. Official NWDA material identifies Ken-to-Betwa transfer and the project as the first National Perspective Plan link under implementation; implementation is not proof of delivered outcomes.

Answer: C. Explanation: Most large Peninsular rivers follow the eastward plateau slope toward Bay of Bengal deltas, while Narmada and Tapi use west-flowing structural troughs and many Ghats streams are short and steep. The other options describe different processes, locations, scales or governance categories.

Q44. Which option avoids the main UPSC trap concerning Peninsular drainage?

A. Official NWDA material identifies Ken-to-Betwa transfer and the project as the first National Perspective Plan link under implementation; implementation is not proof of delivered outcomes. B. A defensible link requires seasonal donor-basin accounting, recipient demand and hydrogeological fit, downstream and environmental-flow needs, conveyance losses and viable local alternatives. C. Ken-Betwa appraisal must include habitat connectivity, downstream flow and sediment, forest and biodiversity effects, displacement, participation, mitigation compliance and long-term monitoring. D. Most large Peninsular rivers follow the eastward plateau slope toward Bay of Bengal deltas, while Narmada and Tapi use west-flowing structural troughs and many Ghats streams are short and steep.

Answer: D. Explanation: Most large Peninsular rivers follow the eastward plateau slope toward Bay of Bengal deltas, while Narmada and Tapi use west-flowing structural troughs and many Ghats streams are short and steep. The other options describe different processes, locations, scales or governance categories.

Q45. Which statement correctly explains Named river map hooks?

A. The named Ganga begins at Devprayag after the Bhagirathi-Alaknanda confluence; Godavari rises near Trimbakeshwar, Krishna near Mahabaleshwar, Kaveri at Talakaveri and Narmada at Amarkantak. B. Channel aggradation, migration, floodplain occupation, embankment confinement and drainage obstruction convert a natural inundation process into differentiated exposure and disaster risk. C. A defensible link requires seasonal donor-basin accounting, recipient demand and hydrogeological fit, downstream and environmental-flow needs, conveyance losses and viable local alternatives. D. Delta behaviour depends on basin sediment supply, wave-tidal energy, subsidence, relative sea level and distributary behaviour; equal sea-level forcing can produce different trajectories.

Answer: A. Explanation: The named Ganga begins at Devprayag after the Bhagirathi-Alaknanda confluence; Godavari rises near Trimbakeshwar, Krishna near Mahabaleshwar, Kaveri at Talakaveri and Narmada at Amarkantak. The other options describe different processes, locations, scales or governance categories.

Q46. Which option is the safest spatial interpretation of Named river map hooks?

A. A defensible link requires seasonal donor-basin accounting, recipient demand and hydrogeological fit, downstream and environmental-flow needs, conveyance losses and viable local alternatives. B. The named Ganga begins at Devprayag after the Bhagirathi-Alaknanda confluence; Godavari rises near Trimbakeshwar, Krishna near Mahabaleshwar, Kaveri at Talakaveri and Narmada at Amarkantak. C. Official NWDA material identifies Ken-to-Betwa transfer and the project as the first National Perspective Plan link under implementation; implementation is not proof of delivered outcomes. D. Channel aggradation, migration, floodplain occupation, embankment confinement and drainage obstruction convert a natural inundation process into differentiated exposure and disaster risk.

Answer: B. Explanation: The named Ganga begins at Devprayag after the Bhagirathi-Alaknanda confluence; Godavari rises near Trimbakeshwar, Krishna near Mahabaleshwar, Kaveri at Talakaveri and Narmada at Amarkantak. The other options describe different processes, locations, scales or governance categories.

Q47. Which statement preserves the process boundary for Named river map hooks?

A. Official NWDA material identifies Ken-to-Betwa transfer and the project as the first National Perspective Plan link under implementation; implementation is not proof of delivered outcomes. B. Ken-Betwa appraisal must include habitat connectivity, downstream flow and sediment, forest and biodiversity effects, displacement, participation, mitigation compliance and long-term monitoring. C. The named Ganga begins at Devprayag after the Bhagirathi-Alaknanda confluence; Godavari rises near Trimbakeshwar, Krishna near Mahabaleshwar, Kaveri at Talakaveri and Narmada at Amarkantak. D. A defensible link requires seasonal donor-basin accounting, recipient demand and hydrogeological fit, downstream and environmental-flow needs, conveyance losses and viable local alternatives.

Answer: C. Explanation: The named Ganga begins at Devprayag after the Bhagirathi-Alaknanda confluence; Godavari rises near Trimbakeshwar, Krishna near Mahabaleshwar, Kaveri at Talakaveri and Narmada at Amarkantak. The other options describe different processes, locations, scales or governance categories.

Q48. Which option avoids the main UPSC trap concerning Named river map hooks?

A. A link moderates floods only with storage space, timing and operating rules; navigation separately requires dependable depth, locks, terminals and sediment management. B. Ken-Betwa appraisal must include habitat connectivity, downstream flow and sediment, forest and biodiversity effects, displacement, participation, mitigation compliance and long-term monitoring. C. Official NWDA material identifies Ken-to-Betwa transfer and the project as the first National Perspective Plan link under implementation; implementation is not proof of delivered outcomes. D. The named Ganga begins at Devprayag after the Bhagirathi-Alaknanda confluence; Godavari rises near Trimbakeshwar, Krishna near Mahabaleshwar, Kaveri at Talakaveri and Narmada at Amarkantak.

Answer: D. Explanation: The named Ganga begins at Devprayag after the Bhagirathi-Alaknanda confluence; Godavari rises near Trimbakeshwar, Krishna near Mahabaleshwar, Kaveri at Talakaveri and Narmada at Amarkantak. The other options describe different processes, locations, scales or governance categories.

Q49. Which statement correctly explains Sediment-budget comparison?

A. Delta behaviour depends on basin sediment supply, wave-tidal energy, subsidence, relative sea level and distributary behaviour; equal sea-level forcing can produce different trajectories. B. Channel aggradation, migration, floodplain occupation, embankment confinement and drainage obstruction convert a natural inundation process into differentiated exposure and disaster risk. C. A defensible link requires seasonal donor-basin accounting, recipient demand and hydrogeological fit, downstream and environmental-flow needs, conveyance losses and viable local alternatives. D. Official NWDA material identifies Ken-to-Betwa transfer and the project as the first National Perspective Plan link under implementation; implementation is not proof of delivered outcomes.

Answer: A. Explanation: Delta behaviour depends on basin sediment supply, wave-tidal energy, subsidence, relative sea level and distributary behaviour; equal sea-level forcing can produce different trajectories. The other options describe different processes, locations, scales or governance categories.

Q50. Which option is the safest spatial interpretation of Sediment-budget comparison?

A. Ken-Betwa appraisal must include habitat connectivity, downstream flow and sediment, forest and biodiversity effects, displacement, participation, mitigation compliance and long-term monitoring. B. Delta behaviour depends on basin sediment supply, wave-tidal energy, subsidence, relative sea level and distributary behaviour; equal sea-level forcing can produce different trajectories. C. A defensible link requires seasonal donor-basin accounting, recipient demand and hydrogeological fit, downstream and environmental-flow needs, conveyance losses and viable local alternatives. D. Official NWDA material identifies Ken-to-Betwa transfer and the project as the first National Perspective Plan link under implementation; implementation is not proof of delivered outcomes.

Answer: B. Explanation: Delta behaviour depends on basin sediment supply, wave-tidal energy, subsidence, relative sea level and distributary behaviour; equal sea-level forcing can produce different trajectories. The other options describe different processes, locations, scales or governance categories.

Q51. Which statement preserves the process boundary for Sediment-budget comparison?

A. Official NWDA material identifies Ken-to-Betwa transfer and the project as the first National Perspective Plan link under implementation; implementation is not proof of delivered outcomes. B. Ken-Betwa appraisal must include habitat connectivity, downstream flow and sediment, forest and biodiversity effects, displacement, participation, mitigation compliance and long-term monitoring. C. Delta behaviour depends on basin sediment supply, wave-tidal energy, subsidence, relative sea level and distributary behaviour; equal sea-level forcing can produce different trajectories. D. A link moderates floods only with storage space, timing and operating rules; navigation separately requires dependable depth, locks, terminals and sediment management.

Answer: C. Explanation: Delta behaviour depends on basin sediment supply, wave-tidal energy, subsidence, relative sea level and distributary behaviour; equal sea-level forcing can produce different trajectories. The other options describe different processes, locations, scales or governance categories.

Q52. Which option avoids the main UPSC trap concerning Sediment-budget comparison?

A. A link moderates floods only with storage space, timing and operating rules; navigation separately requires dependable depth, locks, terminals and sediment management. B. The routed direct Mains demand is 2020 GS-I on interlinking for droughts, floods and navigation; 2021 Prelims routes test Indus tributaries and rivers rising in the Eastern Ghats. C. Ken-Betwa appraisal must include habitat connectivity, downstream flow and sediment, forest and biodiversity effects, displacement, participation, mitigation compliance and long-term monitoring. D. Delta behaviour depends on basin sediment supply, wave-tidal energy, subsidence, relative sea level and distributary behaviour; equal sea-level forcing can produce different trajectories.

Answer: D. Explanation: Delta behaviour depends on basin sediment supply, wave-tidal energy, subsidence, relative sea level and distributary behaviour; equal sea-level forcing can produce different trajectories. The other options describe different processes, locations, scales or governance categories.

Q53. Which statement correctly explains Floodplain risk chain?

A. Channel aggradation, migration, floodplain occupation, embankment confinement and drainage obstruction convert a natural inundation process into differentiated exposure and disaster risk. B. Ken-Betwa appraisal must include habitat connectivity, downstream flow and sediment, forest and biodiversity effects, displacement, participation, mitigation compliance and long-term monitoring. C. Official NWDA material identifies Ken-to-Betwa transfer and the project as the first National Perspective Plan link under implementation; implementation is not proof of delivered outcomes. D. A defensible link requires seasonal donor-basin accounting, recipient demand and hydrogeological fit, downstream and environmental-flow needs, conveyance losses and viable local alternatives.

Answer: A. Explanation: Channel aggradation, migration, floodplain occupation, embankment confinement and drainage obstruction convert a natural inundation process into differentiated exposure and disaster risk. The other options describe different processes, locations, scales or governance categories.

Q54. Which option is the safest spatial interpretation of Floodplain risk chain?

A. Official NWDA material identifies Ken-to-Betwa transfer and the project as the first National Perspective Plan link under implementation; implementation is not proof of delivered outcomes. B. Channel aggradation, migration, floodplain occupation, embankment confinement and drainage obstruction convert a natural inundation process into differentiated exposure and disaster risk. C. A link moderates floods only with storage space, timing and operating rules; navigation separately requires dependable depth, locks, terminals and sediment management. D. Ken-Betwa appraisal must include habitat connectivity, downstream flow and sediment, forest and biodiversity effects, displacement, participation, mitigation compliance and long-term monitoring.

Answer: B. Explanation: Channel aggradation, migration, floodplain occupation, embankment confinement and drainage obstruction convert a natural inundation process into differentiated exposure and disaster risk. The other options describe different processes, locations, scales or governance categories.

Q55. Which statement preserves the process boundary for Floodplain risk chain?

A. A link moderates floods only with storage space, timing and operating rules; navigation separately requires dependable depth, locks, terminals and sediment management. B. The routed direct Mains demand is 2020 GS-I on interlinking for droughts, floods and navigation; 2021 Prelims routes test Indus tributaries and rivers rising in the Eastern Ghats. C. Channel aggradation, migration, floodplain occupation, embankment confinement and drainage obstruction convert a natural inundation process into differentiated exposure and disaster risk. D. Ken-Betwa appraisal must include habitat connectivity, downstream flow and sediment, forest and biodiversity effects, displacement, participation, mitigation compliance and long-term monitoring.

Answer: C. Explanation: Channel aggradation, migration, floodplain occupation, embankment confinement and drainage obstruction convert a natural inundation process into differentiated exposure and disaster risk. The other options describe different processes, locations, scales or governance categories.

Q56. Which option avoids the main UPSC trap concerning Floodplain risk chain?

A. River planning is a basin-scale, federal and ecological exercise combining demand management, recharge, reuse, reservoir operations, floodplain restoration and selective infrastructure. B. A link moderates floods only with storage space, timing and operating rules; navigation separately requires dependable depth, locks, terminals and sediment management. C. The routed direct Mains demand is 2020 GS-I on interlinking for droughts, floods and navigation; 2021 Prelims routes test Indus tributaries and rivers rising in the Eastern Ghats. D. Channel aggradation, migration, floodplain occupation, embankment confinement and drainage obstruction convert a natural inundation process into differentiated exposure and disaster risk.

Answer: D. Explanation: Channel aggradation, migration, floodplain occupation, embankment confinement and drainage obstruction convert a natural inundation process into differentiated exposure and disaster risk. The other options describe different processes, locations, scales or governance categories.

Q57. Which statement correctly explains Inter-basin transfer test?

A. A defensible link requires seasonal donor-basin accounting, recipient demand and hydrogeological fit, downstream and environmental-flow needs, conveyance losses and viable local alternatives. B. Official NWDA material identifies Ken-to-Betwa transfer and the project as the first National Perspective Plan link under implementation; implementation is not proof of delivered outcomes. C. Ken-Betwa appraisal must include habitat connectivity, downstream flow and sediment, forest and biodiversity effects, displacement, participation, mitigation compliance and long-term monitoring. D. A link moderates floods only with storage space, timing and operating rules; navigation separately requires dependable depth, locks, terminals and sediment management.

Answer: A. Explanation: A defensible link requires seasonal donor-basin accounting, recipient demand and hydrogeological fit, downstream and environmental-flow needs, conveyance losses and viable local alternatives. The other options describe different processes, locations, scales or governance categories.

Q58. Which option is the safest spatial interpretation of Inter-basin transfer test?

A. The routed direct Mains demand is 2020 GS-I on interlinking for droughts, floods and navigation; 2021 Prelims routes test Indus tributaries and rivers rising in the Eastern Ghats. B. A defensible link requires seasonal donor-basin accounting, recipient demand and hydrogeological fit, downstream and environmental-flow needs, conveyance losses and viable local alternatives. C. A link moderates floods only with storage space, timing and operating rules; navigation separately requires dependable depth, locks, terminals and sediment management. D. Ken-Betwa appraisal must include habitat connectivity, downstream flow and sediment, forest and biodiversity effects, displacement, participation, mitigation compliance and long-term monitoring.

Answer: B. Explanation: A defensible link requires seasonal donor-basin accounting, recipient demand and hydrogeological fit, downstream and environmental-flow needs, conveyance losses and viable local alternatives. The other options describe different processes, locations, scales or governance categories.

Q59. Which statement preserves the process boundary for Inter-basin transfer test?

A. The routed direct Mains demand is 2020 GS-I on interlinking for droughts, floods and navigation; 2021 Prelims routes test Indus tributaries and rivers rising in the Eastern Ghats. B. A link moderates floods only with storage space, timing and operating rules; navigation separately requires dependable depth, locks, terminals and sediment management. C. A defensible link requires seasonal donor-basin accounting, recipient demand and hydrogeological fit, downstream and environmental-flow needs, conveyance losses and viable local alternatives. D. River planning is a basin-scale, federal and ecological exercise combining demand management, recharge, reuse, reservoir operations, floodplain restoration and selective infrastructure.

Answer: C. Explanation: A defensible link requires seasonal donor-basin accounting, recipient demand and hydrogeological fit, downstream and environmental-flow needs, conveyance losses and viable local alternatives. The other options describe different processes, locations, scales or governance categories.

Q60. Which option avoids the main UPSC trap concerning Inter-basin transfer test?

A. A drainage basin is the area contributing runoff to a river system; a watershed or divide separates neighbouring basins, while tributaries join a trunk stream before its outlet. B. River planning is a basin-scale, federal and ecological exercise combining demand management, recharge, reuse, reservoir operations, floodplain restoration and selective infrastructure. C. The routed direct Mains demand is 2020 GS-I on interlinking for droughts, floods and navigation; 2021 Prelims routes test Indus tributaries and rivers rising in the Eastern Ghats. D. A defensible link requires seasonal donor-basin accounting, recipient demand and hydrogeological fit, downstream and environmental-flow needs, conveyance losses and viable local alternatives.

Answer: D. Explanation: A defensible link requires seasonal donor-basin accounting, recipient demand and hydrogeological fit, downstream and environmental-flow needs, conveyance losses and viable local alternatives. The other options describe different processes, locations, scales or governance categories.

Q61. Which statement correctly explains Ken-Betwa direction and status?

A. Official NWDA material identifies Ken-to-Betwa transfer and the project as the first National Perspective Plan link under implementation; implementation is not proof of delivered outcomes. B. Ken-Betwa appraisal must include habitat connectivity, downstream flow and sediment, forest and biodiversity effects, displacement, participation, mitigation compliance and long-term monitoring. C. A link moderates floods only with storage space, timing and operating rules; navigation separately requires dependable depth, locks, terminals and sediment management. D. The routed direct Mains demand is 2020 GS-I on interlinking for droughts, floods and navigation; 2021 Prelims routes test Indus tributaries and rivers rising in the Eastern Ghats.

Answer: A. Explanation: Official NWDA material identifies Ken-to-Betwa transfer and the project as the first National Perspective Plan link under implementation; implementation is not proof of delivered outcomes. The other options describe different processes, locations, scales or governance categories.

Q62. Which option is the safest spatial interpretation of Ken-Betwa direction and status?

A. The routed direct Mains demand is 2020 GS-I on interlinking for droughts, floods and navigation; 2021 Prelims routes test Indus tributaries and rivers rising in the Eastern Ghats. B. Official NWDA material identifies Ken-to-Betwa transfer and the project as the first National Perspective Plan link under implementation; implementation is not proof of delivered outcomes. C. River planning is a basin-scale, federal and ecological exercise combining demand management, recharge, reuse, reservoir operations, floodplain restoration and selective infrastructure. D. A link moderates floods only with storage space, timing and operating rules; navigation separately requires dependable depth, locks, terminals and sediment management.

Answer: B. Explanation: Official NWDA material identifies Ken-to-Betwa transfer and the project as the first National Perspective Plan link under implementation; implementation is not proof of delivered outcomes. The other options describe different processes, locations, scales or governance categories.

Q63. Which statement preserves the process boundary for Ken-Betwa direction and status?

A. River planning is a basin-scale, federal and ecological exercise combining demand management, recharge, reuse, reservoir operations, floodplain restoration and selective infrastructure. B. A drainage basin is the area contributing runoff to a river system; a watershed or divide separates neighbouring basins, while tributaries join a trunk stream before its outlet. C. Official NWDA material identifies Ken-to-Betwa transfer and the project as the first National Perspective Plan link under implementation; implementation is not proof of delivered outcomes. D. The routed direct Mains demand is 2020 GS-I on interlinking for droughts, floods and navigation; 2021 Prelims routes test Indus tributaries and rivers rising in the Eastern Ghats.

Answer: C. Explanation: Official NWDA material identifies Ken-to-Betwa transfer and the project as the first National Perspective Plan link under implementation; implementation is not proof of delivered outcomes. The other options describe different processes, locations, scales or governance categories.

Q64. Which option avoids the main UPSC trap concerning Ken-Betwa direction and status?

A. Running water erodes by hydraulic action, corrosion or abrasion, attrition and solution; it transports load by traction, saltation, suspension and solution before deposition when competence falls. B. A drainage basin is the area contributing runoff to a river system; a watershed or divide separates neighbouring basins, while tributaries join a trunk stream before its outlet. C. River planning is a basin-scale, federal and ecological exercise combining demand management, recharge, reuse, reservoir operations, floodplain restoration and selective infrastructure. D. Official NWDA material identifies Ken-to-Betwa transfer and the project as the first National Perspective Plan link under implementation; implementation is not proof of delivered outcomes.

Answer: D. Explanation: Official NWDA material identifies Ken-to-Betwa transfer and the project as the first National Perspective Plan link under implementation; implementation is not proof of delivered outcomes. The other options describe different processes, locations, scales or governance categories.

Q65. Which statement correctly explains Panna landscape boundary?

A. Ken-Betwa appraisal must include habitat connectivity, downstream flow and sediment, forest and biodiversity effects, displacement, participation, mitigation compliance and long-term monitoring. B. A link moderates floods only with storage space, timing and operating rules; navigation separately requires dependable depth, locks, terminals and sediment management. C. The routed direct Mains demand is 2020 GS-I on interlinking for droughts, floods and navigation; 2021 Prelims routes test Indus tributaries and rivers rising in the Eastern Ghats. D. River planning is a basin-scale, federal and ecological exercise combining demand management, recharge, reuse, reservoir operations, floodplain restoration and selective infrastructure.

Answer: A. Explanation: Ken-Betwa appraisal must include habitat connectivity, downstream flow and sediment, forest and biodiversity effects, displacement, participation, mitigation compliance and long-term monitoring. The other options describe different processes, locations, scales or governance categories.

Q66. Which option is the safest spatial interpretation of Panna landscape boundary?

A. The routed direct Mains demand is 2020 GS-I on interlinking for droughts, floods and navigation; 2021 Prelims routes test Indus tributaries and rivers rising in the Eastern Ghats. B. Ken-Betwa appraisal must include habitat connectivity, downstream flow and sediment, forest and biodiversity effects, displacement, participation, mitigation compliance and long-term monitoring. C. A drainage basin is the area contributing runoff to a river system; a watershed or divide separates neighbouring basins, while tributaries join a trunk stream before its outlet. D. River planning is a basin-scale, federal and ecological exercise combining demand management, recharge, reuse, reservoir operations, floodplain restoration and selective infrastructure.

Answer: B. Explanation: Ken-Betwa appraisal must include habitat connectivity, downstream flow and sediment, forest and biodiversity effects, displacement, participation, mitigation compliance and long-term monitoring. The other options describe different processes, locations, scales or governance categories.

Q67. Which statement preserves the process boundary for Panna landscape boundary?

A. A drainage basin is the area contributing runoff to a river system; a watershed or divide separates neighbouring basins, while tributaries join a trunk stream before its outlet. B. Running water erodes by hydraulic action, corrasion or abrasion, attrition and solution; it transports load by traction, saltation, suspension and solution before deposition when competence falls. C. Ken-Betwa appraisal must include habitat connectivity, downstream flow and sediment, forest and biodiversity effects, displacement, participation, mitigation compliance and long-term monitoring. D. River planning is a basin-scale, federal and ecological exercise combining demand management, recharge, reuse, reservoir operations, floodplain restoration and selective infrastructure.

Answer: C. Explanation: Ken-Betwa appraisal must include habitat connectivity, downstream flow and sediment, forest and biodiversity effects, displacement, participation, mitigation compliance and long-term monitoring. The other options describe different processes, locations, scales or governance categories.

Q68. Which option avoids the main UPSC trap concerning Panna landscape boundary?

A. A graded river tends toward a concave-up long profile in dynamic balance with discharge, sediment load, rock resistance and base level; uplift, floods and dams can interrupt the tendency. B. Running water erodes by hydraulic action, corrasion or abrasion, attrition and solution; it transports load by traction, saltation, suspension and solution before deposition when competence falls. C. A drainage basin is the area contributing runoff to a river system; a watershed or divide separates neighbouring basins, while tributaries join a trunk stream before its outlet. D. Ken-Betwa appraisal must include habitat connectivity, downstream flow and sediment, forest and biodiversity effects, displacement, participation, mitigation compliance and long-term monitoring.

Answer: D. Explanation: Ken-Betwa appraisal must include habitat connectivity, downstream flow and sediment, forest and biodiversity effects, displacement, participation, mitigation compliance and long-term monitoring. The other options describe different processes, locations, scales or governance categories.

Q69. Which statement correctly explains Flood-navigation limits?

A. A link moderates floods only with storage space, timing and operating rules; navigation separately requires dependable depth, locks, terminals and sediment management. B. River planning is a basin-scale, federal and ecological exercise combining demand management, recharge, reuse, reservoir operations, floodplain restoration and selective infrastructure. C. A drainage basin is the area contributing runoff to a river system; a watershed or divide separates neighbouring basins, while tributaries join a trunk stream before its outlet. D. The routed direct Mains demand is 2020 GS-I on interlinking for droughts, floods and navigation; 2021 Prelims routes test Indus tributaries and rivers rising in the Eastern Ghats.

Answer: A. Explanation: A link moderates floods only with storage space, timing and operating rules; navigation separately requires dependable depth, locks, terminals and sediment management. The other options describe different processes, locations, scales or governance categories.

Q70. Which option is the safest spatial interpretation of Flood-navigation limits?

A. Running water erodes by hydraulic action, corrosion or abrasion, attrition and solution; it transports load by traction, saltation, suspension and solution before deposition when competence falls. B. A link moderates floods only with storage space, timing and operating rules; navigation separately requires dependable depth, locks, terminals and sediment management. C. River planning is a basin-scale, federal and ecological exercise combining demand management, recharge, reuse, reservoir operations, floodplain restoration and selective infrastructure. D. A drainage basin is the area contributing runoff to a river system; a watershed or divide separates neighbouring basins, while tributaries join a trunk stream before its outlet.

Answer: B. Explanation: A link moderates floods only with storage space, timing and operating rules; navigation separately requires dependable depth, locks, terminals and sediment management. The other options describe different processes, locations, scales or governance categories.

Q71. Which statement preserves the process boundary for Flood-navigation limits?

A. A graded river tends toward a concave-up long profile in dynamic balance with discharge, sediment load, rock resistance and base level; uplift, floods and dams can interrupt the tendency. B. Running water erodes by hydraulic action, corrosion or abrasion, attrition and solution; it transports load by traction, saltation, suspension and solution before deposition when competence falls. C. A link moderates floods only with storage space, timing and operating rules; navigation separately requires dependable depth, locks, terminals and sediment management. D. A drainage basin is the area contributing runoff to a river system; a watershed or divide separates neighbouring basins, while tributaries join a trunk stream before its outlet.

Answer: C. Explanation: A link moderates floods only with storage space, timing and operating rules; navigation separately requires dependable depth, locks, terminals and sediment management. The other options describe different processes, locations, scales or governance categories.

Q72. Which option avoids the main UPSC trap concerning Flood-navigation limits?

A. Steep gradients favour vertical incision, V-shaped valleys, gorges, rapids, waterfalls and interlocking spurs, but local rock and discharge matter more than a rigid stage label. B. Running water erodes by hydraulic action, corrosion or abrasion, attrition and solution; it transports load by traction, saltation, suspension and solution before deposition when competence falls. C. A graded river tends toward a concave-up long profile in dynamic balance with discharge, sediment load, rock resistance and base level; uplift, floods and dams can interrupt the tendency. D. A link moderates floods only with storage space, timing and operating rules; navigation separately requires dependable depth, locks, terminals and sediment management.

Answer: D. Explanation: A link moderates floods only with storage space, timing and operating rules; navigation separately requires dependable depth, locks, terminals and sediment management. The other options describe different processes, locations, scales or governance categories.

Q73. Which statement correctly explains Verified PYQ routes?

A. The routed direct Mains demand is 2020 GS-I on interlinking for droughts, floods and navigation; 2021 Prelims routes test Indus tributaries and rivers rising in the Eastern Ghats. B. River planning is a basin-scale, federal and ecological exercise combining demand management, recharge, reuse, reservoir operations, floodplain restoration and selective infrastructure. C. Running water erodes by hydraulic action, corrasion or abrasion, attrition and solution; it transports load by traction, saltation, suspension and solution before deposition when competence falls. D. A drainage basin is the area contributing runoff to a river system; a watershed or divide separates neighbouring basins, while tributaries join a trunk stream before its outlet.

Answer: A. Explanation: The routed direct Mains demand is 2020 GS-I on interlinking for droughts, floods and navigation; 2021 Prelims routes test Indus tributaries and rivers rising in the Eastern Ghats. The other options describe different processes, locations, scales or governance categories.

Demand decoding: The directive **answer** requires a direct position on "Q73. Which statement correctly explains Verified PYQ routes?", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the physical-geography demand in "Q73. Which statement correctly explains Verified PYQ routes?".

Analytical body:

1. **Claim and named evidence:** Q73. Which statement correctly explains Verified PYQ routes? **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** A. The routed direct Mains demand is 2020 GS-I on interlinking for droughts, floods and navigation; 2021 Prelims routes test Indus tributaries and rivers rising in the Eastern Ghats. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** B. River planning is a basin-scale, federal and ecological exercise combining demand management, recharge, reuse, reservoir operations, floodplain restoration and selective infrastructure. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
4. **Claim and named evidence:** C. Running water erodes by hydraulic action, corrasion or abrasion, attrition and solution; it transports load by traction, saltation, suspension and solution before deposition when competence falls. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
5. **Claim and named evidence:** D. A drainage basin is the area contributing runoff to a river system; a watershed or divide separates neighbouring basins, while tributaries join a trunk stream before its outlet. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The answer must resolve the physical-geography demand in "Q73. Which statement correctly explains Verified PYQ routes?".

Executable exam-length answer / compression plan: For a 15-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise five process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for

the causal limit.

Why this earns marks: The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

How to improve this answer: For "Q73. Which statement correctly explains Verified PYQ routes?", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

Q74. Which option is the safest spatial interpretation of Verified PYQ routes?

A. Running water erodes by hydraulic action, corrosion or abrasion, attrition and solution; it transports load by traction, saltation, suspension and solution before deposition when competence falls. B. The routed direct Mains demand is 2020 GS-I on interlinking for droughts, floods and navigation; 2021 Prelims routes test Indus tributaries and rivers rising in the Eastern Ghats. C. A graded river tends toward a concave-up long profile in dynamic balance with discharge, sediment load, rock resistance and base level; uplift, floods and dams can interrupt the tendency. D. A drainage basin is the area contributing runoff to a river system; a watershed or divide separates neighbouring basins, while tributaries join a trunk stream before its outlet.

Answer: B. Explanation: The routed direct Mains demand is 2020 GS-I on interlinking for droughts, floods and navigation; 2021 Prelims routes test Indus tributaries and rivers rising in the Eastern Ghats. The other options describe different processes, locations, scales or governance categories.

Demand decoding: Treat "Q74. Which option is the safest spatial interpretation of Verified PYQ routes?" as a process, location, scale, terminology and status problem. Verify every statement independently.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the physical-geography demand in "Q74. Which option is the safest spatial interpretation of Verified PYQ routes?".

Analytical body:

- 1. Claim and named evidence:** Q74. Which option is the safest spatial interpretation of Verified PYQ routes?
Analysis: Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
- 2. Claim and named evidence:** A. Running water erodes by hydraulic action, corrosion or abrasion, attrition and solution; it transports load by traction, saltation, suspension and solution before deposition when competence falls.
Analysis: Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
- 3. Claim and named evidence:** B. The routed direct Mains demand is 2020 GS-I on interlinking for droughts, floods and navigation; 2021 Prelims routes test Indus tributaries and rivers rising in the Eastern Ghats. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
- 4. Claim and named evidence:** C. A graded river tends toward a concave-up long profile in dynamic balance with discharge, sediment load, rock resistance and base level; uplift, floods and dams can interrupt the tendency.
Analysis: Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
- 5. Claim and named evidence:** D. A drainage basin is the area contributing runoff to a river system; a watershed or divide separates neighbouring basins, while tributaries join a trunk stream before its outlet. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The answer must resolve the physical-geography demand in "Q74. Which option is the safest spatial interpretation of Verified PYQ routes?".

Executable exam-length answer / compression plan: Fix the phenomenon and spatial setting; trace the controlling mechanism; test direction, sequence, threshold and India/world anchor; eliminate the closest term, map or causation distractor.

Why this earns marks: It preserves exact process-to-pattern reasoning and prevents a familiar place-name or correlation from substituting for causation.

How to improve this answer: For "Q74. Which option is the safest spatial interpretation of Verified PYQ routes?", state why the nearest distractor fails on process, direction, scale, location, terminology, date or official status.

Q75. Which statement preserves the process boundary for Verified PYQ routes?

A. Running water erodes by hydraulic action, corrosion or abrasion, attrition and solution; it transports load by traction, saltation, suspension and solution before deposition when competence falls. B. A graded river tends toward a concave-up long profile in dynamic balance with discharge, sediment load, rock resistance and base level; uplift, floods and dams can interrupt the tendency. C. The routed direct Mains demand is 2020 GS-I on interlinking for droughts, floods and navigation; 2021 Prelims routes test Indus tributaries and rivers rising in the Eastern Ghats. D. Steep gradients favour vertical incision, V-shaped valleys, gorges, rapids, waterfalls and interlocking spurs, but local rock and discharge matter more than a rigid stage label.

Answer: C. Explanation: The routed direct Mains demand is 2020 GS-I on interlinking for droughts, floods and navigation; 2021 Prelims routes test Indus tributaries and rivers rising in the Eastern Ghats. The other options describe different processes, locations, scales or governance categories.

Demand decoding: The directive **answer** requires a direct position on "Q75. Which statement preserves the process boundary for Verified PYQ routes?", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the physical-geography demand in "Q75. Which statement preserves the process boundary for Verified PYQ routes?".

Analytical body:

- 1. Claim and named evidence:** Q75. Which statement preserves the process boundary for Verified PYQ routes?
Analysis: Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
- 2. Claim and named evidence:** A. Running water erodes by hydraulic action, corrosion or abrasion, attrition and solution; it transports load by traction, saltation, suspension and solution before deposition when competence falls.
Analysis: Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
- 3. Claim and named evidence:** B. A graded river tends toward a concave-up long profile in dynamic balance with discharge, sediment load, rock resistance and base level; uplift, floods and dams can interrupt the tendency.
Analysis: Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
- 4. Claim and named evidence:** C. The routed direct Mains demand is 2020 GS-I on interlinking for droughts, floods and navigation; 2021 Prelims routes test Indus tributaries and rivers rising in the Eastern Ghats. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

5. **Claim and named evidence:** D. Steep gradients favour vertical incision, V-shaped valleys, gorges, rapids, waterfalls and interlocking spurs, but local rock and discharge matter more than a rigid stage label. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The answer must resolve the physical-geography demand in "Q75. Which statement preserves the process boundary for Verified PYQ routes?".

Executable exam-length answer / compression plan: For a 15-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise five process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

Why this earns marks: The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

How to improve this answer: For "Q75. Which statement preserves the process boundary for Verified PYQ routes?", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

Q76. Which option avoids the main UPSC trap concerning Verified PYQ routes?

A. Steep gradients favour vertical incision, V-shaped valleys, gorges, rapids, waterfalls and interlocking spurs, but local rock and discharge matter more than a rigid stage label. B. A graded river tends toward a concave-up long profile in dynamic balance with discharge, sediment load, rock resistance and base level; uplift, floods and dams can interrupt the tendency. C. Helicoidal flow links outer-bank erosion to inner-bank deposition; neck cutoff can form an ox-bow lake, while repeated overbank deposition builds floodplains and natural levees. D. The routed direct Mains demand is 2020 GS-I on interlinking for droughts, floods and navigation; 2021 Prelims routes test Indus tributaries and rivers rising in the Eastern Ghats.

Answer: D. Explanation: The routed direct Mains demand is 2020 GS-I on interlinking for droughts, floods and navigation; 2021 Prelims routes test Indus tributaries and rivers rising in the Eastern Ghats. The other options describe different processes, locations, scales or governance categories.

Q77. Which statement correctly explains Qualified basin governance?

A. River planning is a basin-scale, federal and ecological exercise combining demand management, recharge, reuse, reservoir operations, floodplain restoration and selective infrastructure. B. A graded river tends toward a concave-up long profile in dynamic balance with discharge, sediment load, rock resistance and base level; uplift, floods and dams can interrupt the tendency. C. Running water erodes by hydraulic action, corrasion or abrasion, attrition and solution; it transports load by traction, saltation, suspension and solution before deposition when competence falls. D. A drainage basin is the area contributing runoff to a river system; a watershed or divide separates neighbouring basins, while tributaries join a trunk stream before its outlet.

Answer: A. Explanation: River planning is a basin-scale, federal and ecological exercise combining demand management, recharge, reuse, reservoir operations, floodplain restoration and selective infrastructure. The other options describe different processes, locations, scales or governance categories.

Q78. Which option is the safest spatial interpretation of Qualified basin governance?

A. Running water erodes by hydraulic action, corrosion or abrasion, attrition and solution; it transports load by traction, saltation, suspension and solution before deposition when competence falls. B. River planning is a basin-scale, federal and ecological exercise combining demand management, recharge, reuse, reservoir operations, floodplain restoration and selective infrastructure. C. A graded river tends toward a concave-up long profile in dynamic balance with discharge, sediment load, rock resistance and base level; uplift, floods and dams can interrupt the tendency. D. Steep gradients favour vertical incision, V-shaped valleys, gorges, rapids, waterfalls and interlocking spurs, but local rock and discharge matter more than a rigid stage label.

Answer: B. Explanation: River planning is a basin-scale, federal and ecological exercise combining demand management, recharge, reuse, reservoir operations, floodplain restoration and selective infrastructure. The other options describe different processes, locations, scales or governance categories.

Q79. Which statement preserves the process boundary for Qualified basin governance?

A. Steep gradients favour vertical incision, V-shaped valleys, gorges, rapids, waterfalls and interlocking spurs, but local rock and discharge matter more than a rigid stage label. B. A graded river tends toward a concave-up long profile in dynamic balance with discharge, sediment load, rock resistance and base level; uplift, floods and dams can interrupt the tendency. C. River planning is a basin-scale, federal and ecological exercise combining demand management, recharge, reuse, reservoir operations, floodplain restoration and selective infrastructure. D. Helicoidal flow links outer-bank erosion to inner-bank deposition; neck cutoff can form an ox-bow lake, while repeated overbank deposition builds floodplains and natural levees.

Answer: C. Explanation: River planning is a basin-scale, federal and ecological exercise combining demand management, recharge, reuse, reservoir operations, floodplain restoration and selective infrastructure. The other options describe different processes, locations, scales or governance categories.

Q80. Which option avoids the main UPSC trap concerning Qualified basin governance?

A. Helicoidal flow links outer-bank erosion to inner-bank deposition; neck cutoff can form an ox-bow lake, while repeated overbank deposition builds floodplains and natural levees. B. A delta develops where sediment accumulation exceeds marine removal and subsidence, whereas an estuary is a drowned or tidally scoured river mouth; estuarine is not a delta shape. C. Steep gradients favour vertical incision, V-shaped valleys, gorges, rapids, waterfalls and interlocking spurs, but local rock and discharge matter more than a rigid stage label. D. River planning is a basin-scale, federal and ecological exercise combining demand management, recharge, reuse, reservoir operations, floodplain restoration and selective infrastructure.

Answer: D. Explanation: River planning is a basin-scale, federal and ecological exercise combining demand management, recharge, reuse, reservoir operations, floodplain restoration and selective infrastructure. The other options describe different processes, locations, scales or governance categories.

PYQS AND ANSWER PRACTICE

Demand decoding: Treat "Q76. Which option avoids the main UPSC trap concerning Verified PYQ routes?" as a process, location, scale, terminology and status problem. Verify every statement independently.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the physical-geography demand in "Q76. Which option avoids the main UPSC trap concerning Verified PYQ routes?".

Analytical body:

- Claim and named evidence:** Q76. Which option avoids the main UPSC trap concerning Verified PYQ routes?
Analysis: Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
- Claim and named evidence:** A. Steep gradients favour vertical incision, V-shaped valleys, gorges, rapids, waterfalls and interlocking spurs, but local rock and discharge matter more than a rigid stage label. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or

observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

3. **Claim and named evidence:** B. A graded river tends toward a concave-up long profile in dynamic balance with discharge, sediment load, rock resistance and base level; uplift, floods and dams can interrupt the tendency. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
4. **Claim and named evidence:** C. Helicoidal flow links outer-bank erosion to inner-bank deposition; neck cutoff can form an ox-bow lake, while repeated overbank deposition builds floodplains and natural levees. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
5. **Claim and named evidence:** D. The routed direct Mains demand is 2020 GS-I on interlinking for droughts, floods and navigation; 2021 Prelims routes test Indus tributaries and rivers rising in the Eastern Ghats. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The answer must resolve the physical-geography demand in "Q76. Which option avoids the main UPSC trap concerning Verified PYQ routes?".

Executable exam-length answer / compression plan: Fix the phenomenon and spatial setting; trace the controlling mechanism; test direction, sequence, threshold and India/world anchor; eliminate the closest term, map or causation distractor.

Why this earns marks: It preserves exact process-to-pattern reasoning and prevents a familiar place-name or correlation from substituting for causation.

How to improve this answer: For "Q76. Which option avoids the main UPSC trap concerning Verified PYQ routes?", state why the nearest distractor fails on process, direction, scale, location, terminology, date or official status.

VERIFIED PYQ OWNERSHIP AUDIT

Verified routing retains the 2020 GS-I interlinking demand and two 2021 Prelims river-map demands. The local routing ledgers do not supply official keys for those objective items; no answer letter is invented.

OWNER PYQ LEDGER EXTRACTS

2026 PYQ Integration

Status: 2026 question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: _PYQ-ROUTING-PRELIMS-2026.md. **Answer-key rule:** The 2026 Prelims and CSAT Set-A keys held locally are **provisional**; no option or answer is recorded or inferred in this integration.

- **Year represented:** 2026
- **Paper(s):** Prelims GS-I
- **Routed question demands:** 2

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2026	Prelims GS-I	11	Pleistocene shifts in Yamuna, Sutlej, Ganga, and Indus drainage	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (Ans - ■2026 - ■GS1 - ■Provis ional); key is provisional - no answer letter recorded or inferred here	Cover the named fact/concept and its likely statement-level distinctions.
2026	Prelims GS-I	25	Indian subcontinent river identified through antecedence, course, origin, and distributaries	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (Ans - ■2026 - ■GS1 - ■Provis ional); key is provisional - no answer letter recorded or inferred here	Cover the named fact/concept and its likely statement-level distinctions.

What this owner must now support

- Pleistocene shifts in Yamuna, Sutlej, Ganga, and Indus drainage
- Indian subcontinent river identified through antecedence, course, origin, and distributaries

This block integrates the 2026 examinable demand and paper metadata. It is kept separate from the 2018-2023 and 2024-2025 blocks and does not convert a provisionally-keyed, answer-free objective question into a solved answer.

Recent PYQ Integration (2024-2025)

Status: 2024-2025 question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: _PYQ-ROUTING-PRELIMS-2024-2025.md. **Answer-key rule:** The official 2024-2025 Prelims Set-A keys are present in the repository and CSAT Set-A keys are supplied; even so, no option or answer is recorded or inferred in this integration.

- **Years represented:** 2024
- **Paper(s):** Prelims GS-I
- **Routed question demands:** 1

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2024	Prelims GS-I	9	Waterfall-region-river matching (Dhuandhar, Hundru, Gersoppa)	Objective question; official Set-A key available locally, answer not inferred	Key available locally (official Set-A answer key present); answer not recorded here	Cover the named fact/concept and its likely statement-level distinctions.

What this owner must now support

- Waterfall-region-river matching (Dhuandhar, Hundru, Gersoppa)

This block integrates the 2024-2025 examinable demand and paper metadata. It is kept separate from the 2018-2023 block and does not convert an unkeyed/answer-free objective question into a solved answer.

Historical PYQ Integration (2018-2023)

Status: Question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: _PYQ-ROUTING-PRELIMS-2018-2023.md. **Answer-key rule:** The official 2018-2023 Prelims/CSAT keys are not held locally; no option or answer has been inferred.

- **Years represented:** 2021, 2022
- **Paper(s):** Prelims GS-I
- **Routed question demands:** 3

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2021	Prelims GS-I	53	Indus river system tributaries direct junction	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2021	Prelims GS-I	55	Rivers rising from Eastern Ghats identification	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2022	Prelims GS-I	24	Gandikota canyon formation by South Indian river	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.

What this owner must now support

- Indus river system tributaries direct junction
- Rivers rising from Eastern Ghats identification
- Gandikota canyon formation by South Indian river

The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

Historical PYQ Integration (2018-2023)

Status: Question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: _PYQ-ROUTING-MAINS-GS1-GS2-ESSAY-2018-2023.md.

- **Years represented:** 2020
- **Paper(s):** GS-I
- **Routed question demands:** 1

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2020	GS-I	14	Interlinking of rivers for droughts floods and navigation	Critically examine · 15 marks · 250 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.

What this owner must now support

- Interlinking of rivers for droughts floods and navigation

The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

PYQ DEMAND CARD 1 - 2020 GS-I

Demand: The interlinking of rivers can provide viable solutions to droughts, floods and interrupted navigation. Critically examine. (250 words)

Status: Verified routed direct Mains demand.

Model solution: Separate the three claims. Transfer may relocate seasonal water; flood moderation needs storage and operating rules; navigation needs depth and infrastructure. Use Ken-to-Betwa implementation as dated evidence, then test donor balance, downstream and environmental flows, Panna landscape effects, social costs, federal coordination and cheaper local alternatives before a conditional verdict.

Demand decoding: The directive **answer** requires a direct position on "PYQ DEMAND CARD 1 - 2020 GS-I", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the physical-geography demand in "PYQ DEMAND CARD 1 - 2020 GS-I".

Analytical body:

1. **Claim and named evidence:** Demand: The interlinking of rivers can provide viable solutions to droughts, floods and interrupted navigation. Critically examine. (250 words) **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive.

Qualification: State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The answer must resolve the physical-geography demand in "PYQ DEMAND CARD 1 - 2020 GS-I".

Executable exam-length answer / compression plan: For a 15-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise five process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

Why this earns marks: The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

How to improve this answer: For "PYQ DEMAND CARD 1 - 2020 GS-I", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

PYQ DEMAND CARD 2 - 2021 Prelims GS-I

Demand: Indus river-system tributaries and their direct junctions.

Status: Verified routed objective demand; official key unavailable in the ledger.

Model solution: Solve by tracing direct confluences on a basin sketch rather than by proximity. Preserve the official option order when using the paper; do not infer an answer letter from this neutral routing summary.

Demand decoding: Treat "PYQ DEMAND CARD 2 - 2021 Prelims GS-I" as a process, location, scale, terminology and status problem. Verify every statement independently.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the physical-geography demand in "PYQ DEMAND CARD 2 - 2021 Prelims GS-I".

Analytical body:

1. **Claim and named evidence:** Demand: Indus river-system tributaries and their direct junctions. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** Status: Verified routed objective demand; official key unavailable in the ledger. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** Model solution: Solve by tracing direct confluences on a basin sketch rather than by proximity. Preserve the official option order when using the paper; do not infer an answer letter from this neutral routing summary. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and

connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The answer must resolve the physical-geography demand in "PYQ DEMAND CARD 2 - 2021 Prelims GS-I".

Executable exam-length answer / compression plan: Fix the phenomenon and spatial setting; trace the controlling mechanism; test direction, sequence, threshold and India/world anchor; eliminate the closest term, map or causation distractor.

Why this earns marks: It preserves exact process-to-pattern reasoning and prevents a familiar place-name or correlation from substituting for causation.

How to improve this answer: For "PYQ DEMAND CARD 2 - 2021 Prelims GS-I", state why the nearest distractor fails on process, direction, scale, location, terminology, date or official status.

PYQ DEMAND CARD 3 - 2021 Prelims GS-I

Demand: Identification of rivers rising in the Eastern Ghats.

Status: Verified routed objective demand; official key unavailable in the ledger.

Model solution: Separate origin from course: several rivers cross or drain the Eastern Ghats without rising there. Use source-location mapping and do not manufacture an official key.

Demand decoding: Treat "PYQ DEMAND CARD 3 - 2021 Prelims GS-I" as a process, location, scale, terminology and status problem. Verify every statement independently.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the physical-geography demand in "PYQ DEMAND CARD 3 - 2021 Prelims GS-I".

Analytical body:

1. **Claim and named evidence:** Demand: Identification of rivers rising in the Eastern Ghats. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** Status: Verified routed objective demand; official key unavailable in the ledger. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** Model solution: Separate origin from course: several rivers cross or drain the Eastern Ghats without rising there. Use source-location mapping and do not manufacture an official key. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The answer must resolve the physical-geography demand in "PYQ DEMAND CARD 3 - 2021 Prelims GS-I".

Executable exam-length answer / compression plan: Fix the phenomenon and spatial setting; trace the controlling mechanism; test direction, sequence, threshold and India/world anchor; eliminate the closest term, map or causation distractor.

Why this earns marks: It preserves exact process-to-pattern reasoning and prevents a familiar place-name or correlation from substituting for causation.

How to improve this answer: For "PYQ DEMAND CARD 3 - 2021 Prelims GS-I", state why the nearest distractor fails on process, direction, scale, location, terminology, date or official status.

ORIGINAL MAINS 1 - 10 MARKS

Question: Explain how a river's energy-load relation creates a sequence of erosional and depositional landforms. Answer in about 150 words.

Model thesis: Fluvial form follows changing energy relative to load, modified by rock, discharge, base level and tectonic or human interruption.

Claim -> named evidence -> analysis -> qualification:

- Running water erodes by hydraulic action, corrosion or abrasion, attrition and solution; it transports load by traction, saltation, suspension and solution before deposition when competence falls.
- A graded river tends toward a concave-up long profile in dynamic balance with discharge, sediment load, rock resistance and base level; uplift, floods and dams can interrupt the tendency.
- Steep gradients favour vertical incision, V-shaped valleys, gorges, rapids, waterfalls and interlocking spurs, but local rock and discharge matter more than a rigid stage label.
- Helicoidal flow links outer-bank erosion to inner-bank deposition; neck cutoff can form an ox-bow lake, while repeated overbank deposition builds floodplains and natural levees.

Qualified conclusion: Fluvial form follows changing energy relative to load, modified by rock, discharge, base level and tectonic or human interruption.

Demand decoding: The directive **explain** requires a direct position on "Explain how a river's energy-load relation creates a sequence of erosional and depositional...", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: Fluvial form follows changing energy relative to load, modified by rock, discharge, base level and tectonic or human interruption.

Analytical body:

1. **Claim and named evidence:** Running water erodes by hydraulic action, corrosion or abrasion, attrition and solution; it transports load by traction, saltation, suspension and solution before deposition when competence falls. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** A graded river tends toward a concave-up long profile in dynamic balance with discharge, sediment load, rock resistance and base level; uplift, floods and dams can interrupt the tendency. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** Steep gradients favour vertical incision, V-shaped valleys, gorges, rapids, waterfalls and interlocking spurs, but local rock and discharge matter more than a rigid stage label. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
4. **Claim and named evidence:** Helicoidal flow links outer-bank erosion to inner-bank deposition; neck cutoff can form an ox-bow lake, while repeated overbank deposition builds floodplains and natural levees. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: Fluvial form follows changing energy relative to load, modified by rock, discharge, base level and tectonic or human interruption.

Executable exam-length answer / compression plan: For a 10-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise three process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

Why this earns marks: The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

How to improve this answer: For "Explain how a river's energy-load relation creates a sequence of erosional and depositional...", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

ORIGINAL MAINS 2 - 10 MARKS

Question: Differentiate antecedent drainage, superimposed drainage and river capture with diagnostic evidence. Answer in about 150 words.

Model thesis: The three concepts differ by timing and mechanism: persistence through uplift, inheritance from cover, and diversion by headward erosion.

Claim -> named evidence -> analysis -> qualification:

- An antecedent river maintains a course across rising relief by incision, while a superimposed river inherits a pattern from a removed cover and cuts into unrelated underlying structure.
- Headward erosion can divert a neighbouring stream, producing an elbow of capture, wind gap and misfit reach; capture is a drainage-evolution process, not ordinary tributary joining.

Qualified conclusion: The three concepts differ by timing and mechanism: persistence through uplift, inheritance from cover, and diversion by headward erosion.

Demand decoding: The directive **answer** requires a direct position on "Differentiate antecedent drainage, superimposed drainage and river capture with diagnostic...", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: The three concepts differ by timing and mechanism: persistence through uplift, inheritance from cover, and diversion by headward erosion.

Analytical body:

1. **Claim and named evidence:** An antecedent river maintains a course across rising relief by incision, while a superimposed river inherits a pattern from a removed cover and cuts into unrelated underlying structure. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** Headward erosion can divert a neighbouring stream, producing an elbow of capture, wind gap and misfit reach; capture is a drainage-evolution process, not ordinary tributary joining. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The three concepts differ by timing and mechanism: persistence through uplift, inheritance from cover, and diversion by headward erosion.

Executable exam-length answer / compression plan: For a 10-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise three process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for

the causal limit.

Why this earns marks: The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

How to improve this answer: For "Differentiate antecedent drainage, superimposed drainage and river capture with diagnostic...", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

ORIGINAL MAINS 3 - 15 MARKS

Question: Compare Himalayan and Peninsular drainage on regime, structure, valley form and mouth type. Answer in about 250 words.

Model thesis: Relief, geological age, water sources and structural controls explain the contrast, while west-flowing trough rivers and regulated reaches qualify stereotypes.

Claim -> named evidence -> analysis -> qualification:

- The Indus, Ganga and Brahmaputra systems cross high relief and international boundaries; their regimes combine rain, snow or glacier melt and groundwater in varying proportions.
- Most large Peninsular rivers follow the eastward plateau slope toward Bay of Bengal deltas, while Narmada and Tapi use west-flowing structural troughs and many Ghats streams are short and steep.
- The named Ganga begins at Devprayag after the Bhagirathi-Alaknanda confluence; Godavari rises near Trimbakeshwar, Krishna near Mahabaleshwar, Kaveri at Talakaveri and Narmada at Amarkantak.

Qualified conclusion: Relief, geological age, water sources and structural controls explain the contrast, while west-flowing trough rivers and regulated reaches qualify stereotypes.

Demand decoding: The directive **compare** requires a direct position on "Compare Himalayan and Peninsular drainage on regime, structure, valley form and mouth type....", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: Relief, geological age, water sources and structural controls explain the contrast, while west-flowing trough rivers and regulated reaches qualify stereotypes.

Analytical body:

1. **Claim and named evidence:** The Indus, Ganga and Brahmaputra systems cross high relief and international boundaries; their regimes combine rain, snow or glacier melt and groundwater in varying proportions. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** Most large Peninsular rivers follow the eastward plateau slope toward Bay of Bengal deltas, while Narmada and Tapi use west-flowing structural troughs and many Ghats streams are short and steep. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** The named Ganga begins at Devprayag after the Bhagirathi-Alaknanda confluence; Godavari rises near Trimbakeshwar, Krishna near Mahabaleshwar, Kaveri at Talakaveri and Narmada at Amarkantak. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: Relief, geological age, water sources and structural controls explain the contrast, while west-flowing trough rivers and regulated reaches qualify stereotypes.

Executable exam-length answer / compression plan: For a 15-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise five process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

Why this earns marks: The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

How to improve this answer: For "Compare Himalayan and Peninsular drainage on regime, structure, valley form and mouth type....", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

ORIGINAL MAINS 4 - 15 MARKS

Question: Assess why deltas facing the same sea-level trend can show different patterns of advance and retreat. Answer in about 250 words.

Model thesis: Sediment budget, marine energy, subsidence and distributary management mediate common sea-level forcing.

Claim -> named evidence -> analysis -> qualification:

- A delta develops where sediment accumulation exceeds marine removal and subsidence, whereas an estuary is a drowned or tidally scoured river mouth; estuarine is not a delta shape.
- Delta behaviour depends on basin sediment supply, wave-tidal energy, subsidence, relative sea level and distributary behaviour; equal sea-level forcing can produce different trajectories.
- Channel aggradation, migration, floodplain occupation, embankment confinement and drainage obstruction convert a natural inundation process into differentiated exposure and disaster risk.

Qualified conclusion: Sediment budget, marine energy, subsidence and distributary management mediate common sea-level forcing.

Demand decoding: The directive **assess** requires a direct position on "Assess why deltas facing the same sea-level trend can show different patterns of advance and...", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: Sediment budget, marine energy, subsidence and distributary management mediate common sea-level forcing.

Analytical body:

1. **Claim and named evidence:** A delta develops where sediment accumulation exceeds marine removal and subsidence, whereas an estuary is a drowned or tidally scoured river mouth; estuarine is not a delta shape.
Analysis: Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** Delta behaviour depends on basin sediment supply, wave-tidal energy, subsidence, relative sea level and distributary behaviour; equal sea-level forcing can produce different trajectories.
Analysis: Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** Channel aggradation, migration, floodplain occupation, embankment confinement and drainage obstruction convert a natural inundation process into differentiated exposure and disaster risk.
Analysis: Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: Sediment budget, marine energy, subsidence and distributary management mediate common sea-level forcing.

Executable exam-length answer / compression plan: For a 15-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise five process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

Why this earns marks: The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

How to improve this answer: For "Assess why deltas facing the same sea-level trend can show different patterns of advance and...", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

ORIGINAL MAINS 5 - 20 MARKS

Question: Critically examine river interlinking as a response to droughts, floods and interrupted navigation. Answer in about 300 words.

Model thesis: Each claimed benefit requires a separate mechanism and test; selective transfer can help, but basin accounting, ecology, federalism and alternatives control viability.

Claim -> named evidence -> analysis -> qualification:

- A defensible link requires seasonal donor-basin accounting, recipient demand and hydrogeological fit, downstream and environmental-flow needs, conveyance losses and viable local alternatives.
- Official NWDA material identifies Ken-to-Betwa transfer and the project as the first National Perspective Plan link under implementation; implementation is not proof of delivered outcomes.
- Ken-Betwa appraisal must include habitat connectivity, downstream flow and sediment, forest and biodiversity effects, displacement, participation, mitigation compliance and long-term monitoring.
- A link moderates floods only with storage space, timing and operating rules; navigation separately requires dependable depth, locks, terminals and sediment management.
- River planning is a basin-scale, federal and ecological exercise combining demand management, recharge, reuse, reservoir operations, floodplain restoration and selective infrastructure.

Qualified conclusion: Each claimed benefit requires a separate mechanism and test; selective transfer can help, but basin accounting, ecology, federalism and alternatives control viability.

Demand decoding: The directive **critically examine** requires a direct position on "Critically examine river interlinking as a response to droughts, floods and interrupted...", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: Each claimed benefit requires a separate mechanism and test; selective transfer can help, but basin accounting, ecology, federalism and alternatives control viability.

Analytical body:

1. **Claim and named evidence:** A defensible link requires seasonal donor-basin accounting, recipient demand and hydrogeological fit, downstream and environmental-flow needs, conveyance losses and viable local alternatives.
Analysis: Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** Official NWDA material identifies Ken-to-Betwa transfer and the project as the first National Perspective Plan link under implementation; implementation is not proof of delivered outcomes.
Analysis: Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

3. **Claim and named evidence:** Ken-Betwa appraisal must include habitat connectivity, downstream flow and sediment, forest and biodiversity effects, displacement, participation, mitigation compliance and long-term monitoring. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
4. **Claim and named evidence:** A link moderates floods only with storage space, timing and operating rules; navigation separately requires dependable depth, locks, terminals and sediment management. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
5. **Claim and named evidence:** River planning is a basin-scale, federal and ecological exercise combining demand management, recharge, reuse, reservoir operations, floodplain restoration and selective infrastructure. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: Each claimed benefit requires a separate mechanism and test; selective transfer can help, but basin accounting, ecology, federalism and alternatives control viability.

Executable exam-length answer / compression plan: For a 20-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise six to eight process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

Why this earns marks: The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

How to improve this answer: For "Critically examine river interlinking as a response to droughts, floods and interrupted...", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

ORIGINAL MAINS 6 - 20 MARKS

Question: Design an integrated basin-governance framework for Indian rivers under climatic and sediment uncertainty. Answer in about 300 words.

Model thesis: Combine hydrological accounting, sediment and floodplain management, demand-side reform, federal coordination, ecological safeguards and adaptive monitoring.

Claim -> named evidence -> analysis -> qualification:

- Delta behaviour depends on basin sediment supply, wave-tidal energy, subsidence, relative sea level and distributary behaviour; equal sea-level forcing can produce different trajectories.
- Channel aggradation, migration, floodplain occupation, embankment confinement and drainage obstruction convert a natural inundation process into differentiated exposure and disaster risk.
- A defensible link requires seasonal donor-basin accounting, recipient demand and hydrogeological fit, downstream and environmental-flow needs, conveyance losses and viable local alternatives.
- Ken-Betwa appraisal must include habitat connectivity, downstream flow and sediment, forest and biodiversity effects, displacement, participation, mitigation compliance and long-term monitoring.
- River planning is a basin-scale, federal and ecological exercise combining demand management, recharge, reuse, reservoir operations, floodplain restoration and selective infrastructure.

Qualified conclusion: Combine hydrological accounting, sediment and floodplain management, demand-side reform, federal coordination, ecological safeguards and adaptive monitoring.

Demand decoding: The directive **answer** requires a direct position on "Design an integrated basin-governance framework for Indian rivers under climatic and sediment...", all clauses and scales, a process chain, spatial

reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: Combine hydrological accounting, sediment and floodplain management, demand-side reform, federal coordination, ecological safeguards and adaptive monitoring.

Analytical body:

1. **Claim and named evidence:** Delta behaviour depends on basin sediment supply, wave-tidal energy, subsidence, relative sea level and distributary behaviour; equal sea-level forcing can produce different trajectories.
Analysis: Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** Channel aggradation, migration, floodplain occupation, embankment confinement and drainage obstruction convert a natural inundation process into differentiated exposure and disaster risk.
Analysis: Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** A defensible link requires seasonal donor-basin accounting, recipient demand and hydrogeological fit, downstream and environmental-flow needs, conveyance losses and viable local alternatives.
Analysis: Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
4. **Claim and named evidence:** Ken-Betwa appraisal must include habitat connectivity, downstream flow and sediment, forest and biodiversity effects, displacement, participation, mitigation compliance and long-term monitoring. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
5. **Claim and named evidence:** River planning is a basin-scale, federal and ecological exercise combining demand management, recharge, reuse, reservoir operations, floodplain restoration and selective infrastructure. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: Combine hydrological accounting, sediment and floodplain management, demand-side reform, federal coordination, ecological safeguards and adaptive monitoring.

Executable exam-length answer / compression plan: For a 20-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise six to eight process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

Why this earns marks: The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

How to improve this answer: For "Design an integrated basin-governance framework for Indian rivers under climatic and sediment...", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.